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PERTRONIC
NETWORK SYSTEM
TECHNICAL MANUAL
AUSTRALIA

Valid for Network System:

Network Card:	**Hardware:**	**v2.07 and greater**
Network Card: Panel PCB	**Firmware:**	**v2.04 (PCBv2.07-2.09), v3.04 (PCBv2.20 and greater)**
Network Card: Supervisor PCB	**Firmware:**	**v2.04 (PCBv2.07-2.09), v3.14 (PCBv2.20 and greater)**
Network Control Unit (NCU)	**Hardware:**	**PCB v1.00 and greater**
	Firmware:	**v1.04 and greater**
Network Display Unit (NDU)	**Hardware:**	**v2.1 and greater**
	Firmware:	**v1.00 and greater**
Network Utilities		**v2.10 and greater**
FISP Programmer		**v3.2.1.7 and greater**
AVR 'Flash' Programmer		**v1.2 and greater**
PonyProg 2000 Flash Programmer		**v2.06 and greater**

Issue 6.2

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Acronyms:

BCO	Bell Cut-Off	- Bell Cut-Off switch terminates the output from the panel BELL relay. Two BCO switches are provided, an External BCO switch for Brigade and Test use and an Internal BCO for Fire Technician use.
AVF	Alarm Verification Facility	- a facility designed to minimise 'nuisance alarms' by ensuring multiple operation of smoke detectors before the alarm is raised.
EOL	End of Line	- an end of line load (10kΩ Resistor) used to monitor the presence and integrity of the detector and sounder circuits.
NDU	Network Display Unit	2 x 40 character LCD with global system controls
NCU	Network Control Unit	Graphics LCD with global system controls and panel monitor function
LED	Light Emitting Diode	
MCP	Manual Call-Point	
OC	Open-circuit	
PCB	Printed Circuit Board	
SC	Short-circuit	

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1. INTRODUCTION

The ***Pertronic Network System*** can connect up to 99 ***Pertronic*** Network Cards via a two-wire high speed RS485 data communication loop connected in a closed ring circuit. The network is fault tolerant so that a single break or short-circuit anywhere in the communication loop will not compromise the operation of the system. The network cards can be connected to F100 (v4.54 or later firmware) or F120 (v3.12 or later firmware) control panels, Network Control Units, Network Display Units, Mini-mimics, LED Mimics or Pager Bases in various combinations.

1.1 Functional Description

1.1.1 Panel to Panel Mapping

An event at one panel may be mapped as a network input to one or more other panels for 'Alarm' indication or to activate system relays or other mappable outputs. For example, a fire zone in 'Alarm' on Panel 3 can be mapped to activate the evacuation sounders and Relay 23 on Loop 2 of Panel 5.

Up to 64 network inputs may be used for ***F100A*** panels, and up to 128 network inputs for ***F120A*** panels and network ***LED mimics***.

Each panel and LED mimic has a **Network Panel** type network card, product code: **NETCARD**.

Network Utilities software is used to configure the network cards with the details of all the network cards present, their functions, the input / output mapping functions for the particular application, and then upload the configuration to the network cards from any network card on the network. It can also download the current configuration from the network to the PC, or verify the network against an existing configuration file.

No other network devices are necessary to perform panel-to-panel mapping.

1.1.2 Network Card

The Network card is the communications device for all network messages and is the only device to be physically connected directly to the network communications loop. All other peripheral devices, such as panels and mimics, are connected to the network via a separate port on the network cards.

There are two types of network card, one for panels and network LED mimics (**Network Panel**), and the other for display units and pager interfaces (**Network NCU**).

Refer to Section 2: 'Network Card' for more details on network cards.

1.1.3 NDU and NCU

The **NDU** (Network Display Unit) and **NCU** (Network Control Unit) are LCD / keyboard displays connected to a **Network NCU** type network card, product code: **NETCD-NCU**.

Each **Network NCU** card can have either one NCU connected, or up to eight (8) NDUs connected. In addition to the display, the card can be configured through the Utility to drive a Pager Base via the on-board RS232 port.

Although the inclusion of an NDU or NCU in a network system is not a specific requirement, it is recommended to provide a site wide view of events.

1.1.3.1 NDU - Network Display Unit:

The NDU displays the status of the network panels by maintaining seven event queues:

Alarm queue
Network Input queue
Pre-alarm queue
Isolate queue
Network Fault queue
Panel Fault queue
System Event queue

The Fire-Fighter Facility operates across the network, allowing devices in the 'Alarm' state to be Acknowledged, Reset or Isolated from the NDU.

It also allows External Bells, Warning System, Door Holder and ACF to be Isolated globally.

The NDU has global control to turn the Warning System ON - this affects all control panels.

The NDU uses a 2-line by 40-character alphanumeric LCD as the primary display, supported by LED indicators identical to those found on the **F100A** and **F120A** control panels.

Refer to Section 3: 'Network Display Unit (NDU)' for more details of the NDU display.

1.1.3.2 NCU - Network Control Unit:

The NCU displays the network status and event queues, and allows the same Fire Fighter Facility functions as the NDU.

The NCU can communicate directly to an **F100A** or **F120A** panel on the network, and operate the panel using the NCU keyboard and LCD display, so that all operator functions of the Fire panel display can be performed as though the user was at the panel itself. Communication is initiated via a menu option

The NCU has a large graphic display providing up to 10 lines of text. The LED indicators are similar to the NDU and control panel, with three (3) additional indicators for Network Status (Alarm, Fault or Isolate), a 'Panel Select' function and four additional switches for 'Walk Test' and 'Lamp Test' with LED indicators plus, 'Enter' and 'Menu' functions.

All NCU Controls bar the Walk Test, Lamp Test and Enter are global, affecting all control panels, unless the display is 'connected' to an individual panel, in which case the commands are relevant to that panel only.

Refer to Section 4 for more details on the NCU display.

1.1.4 Network LED Display

The Network LED Display is an array of up to 128 LED's that can be activated from events around the network. LED activations from network panels can be mapped to activate LEDs of the Network LED Display. This mapping is set up in the Network Utilities software, and is downloaded to the network as part of the program configuration.

The Network LED Display uses a Network Panel type network card, product code: **NETCARD**.

Up to 16 LED Display boards (128 LEDs) can be connected. Either the 12-way Smart LED card (F100PDB12) or the LED Address Controller (LAC485), can be used with 8-way LED Extenders (F100PDB). The 12-way Smart LED card offers global status indicators, which do not require mapping.

Refer to Section 5: 'Network LED Display Operation' for more details on the Network LED Display.

1.1.5 Network Pager

The Network Pager connects to a third party pager system through the RS232 port.

Three pager options are currently supported: AUSTCO, AUSTCO-APR and KEW.

The Network Pager is configured from the Network Utilities software.

The Network Pager uses a Network NCU type network card.

Refer to Section 6: 'Pager Operation' for more details on the Network Pager.

1.1.6 Network Printer

An interface to a serial printer is provided through the supervisory network cards RS232 port (firmware version 3.0 and above only).

The printed network event queues can be selected when configuring the card in the Network Utilities software.

The Network Printer uses a Network NCU type network card.

Refer to Section 7 'PRINTER INTERFACE' for more details on the Network Printer.

1.1.7 Network Utilities

The Network Utilities package is a software application that runs under Windows 98 (SP2), NT, 2000 or XP, (v2.06); Windows XP or Vista (v2.08).

It performs the following network configuration functions:

- Panel identity setup
- Display identity setup
- Pager setup
- Panel to panel / LED Mimic mapping
- Upload the network configuration to the network cards

Download the network configuration from the network to the PC
Verify the network configuration against a saved configuration file
Configuration file management and printout.
Network history event log download (firmware version 3.0 and above only).
Network debug and diagnostic tools (firmware version 2.0 and above only).

Refer to Section 8: ‘Network Configuration Program’ for more details on the Network Configuration Utility.

2. NETWORK CARD

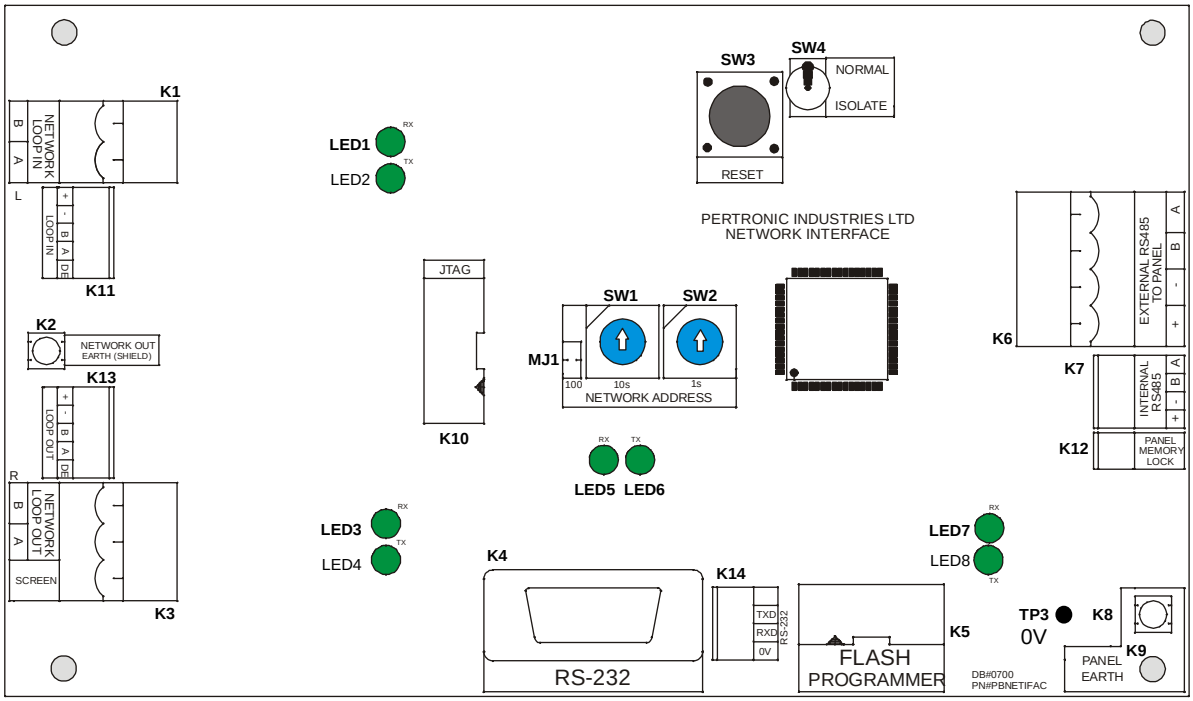


Figure 2.1: Network Card PCB v2.09

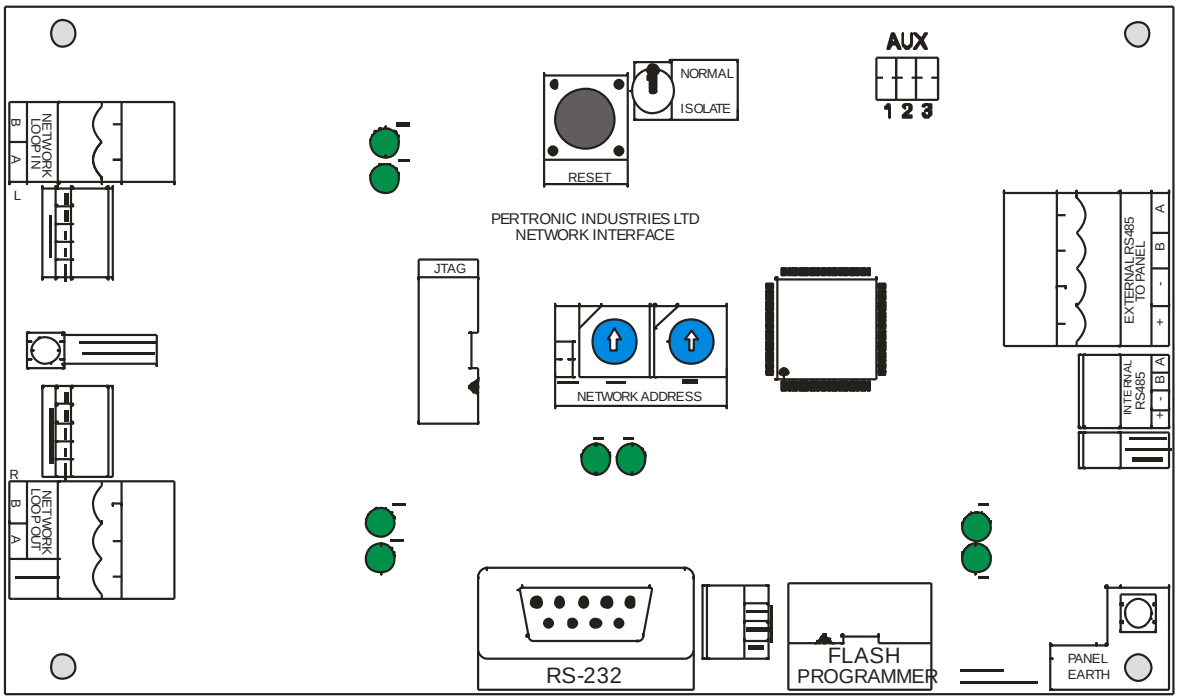


Figure 2.2: Network Card PCB v2.20

2.1 Specification

PCB Dimensions:	166 x 97 x 28	H x W x D mm
Mounting Holes:	152 x 89mm centres	4 x M4 Mounting Holes
Power Supply:	55mA @ 27.4Vdc	- Network Card alone (PCB v2.09)
	75mA @ 27.4Vdc	- Network Card alone (PCB v2.20)
Network Control Unit	+ 18mA	- Backlight and Buzzer OFF
Backlight	+ 7mA	- Backlight ON
Buzzer	+ 15mA	- Buzzer ON

Network Display Unit	+ 6mA	- Backlight and Buzzer OFF
Backlight + LED	+ 32mA	- Backlight and ‘Alarm’ and/or ‘Fault’ LED ON
Buzzer	+ 11mA	- Buzzer ON

Network Circuit: Typically powered from a Pertronic Fire Panel through the RS485 bus.

Two-wire high-speed RS485 data communication loop

Fault tolerant - a single break or short-circuit anywhere in the communication loop will not compromise the operation of the system.

- each link between network cards is individually monitored for communications failure and Earth fault.

RS232 interface to external devices - Network Configuration Utility, Pager, etc.

Flash Programmer interface for firmware upgrades using FISP, PonyProg 2000 or AVR Programmer.

Has a unique Address for each Network card.

Reset switch provides a ‘soft’ reset to the microprocessor

Isolate switch disables communications in both directions between the Control Panel and the Network.

2.2 Firmware

There are two types of Network Card:

Network Panel (NETCARD): for Fire Panels and Network LED Mimics

Network NCU (NETCD-NCU): for Display Units (NCU and NDU) and Pager interfaces.

Version 2.0 firmware is used with v2.07 - v2.09 PCB; version 3.0 firmware is required when using v2.20 PCB. Version 3.0 firmware (v2.20 PCB) is backwards compatible with version 2.0 firmware (v2.09 PCB) if the AUX 1 jumper is fitted; this lowers the baud rate on the network loop communications to match the version 2.0 firmware. Version 1.08 firmware is not compatible with either version 2.0 or 3.0 firmware.

With the exception of mixing version 2.0 and 3.0 firmware on the same network, network firmware on all network cards must be of the same version.

2.3 Connections

Network Cards require a power supply (24Vdc) and the network loop, at least, to function. The RS485 port and/or the RS232 port may also be required, depending on the peripheral equipment attached to the card.

Twisted pair cable (e.g. Pertronic EVACCBL) is recommended for interconnecting Network Cards.

A maximum distance of 1km between network cards is achievable using the listed cables.

2.3.1 Power and RS485

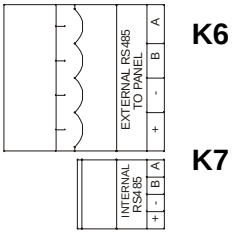


Figure 2.3: RS485

Connectors K6 and K7 connect peripheral equipment to the Network card’s RS485 port.

If mounted within the panel, LED mimics, NDU or NCU displays connect using the internal RS485 connection. If mounted externally, the External connection is used. Network cards are typically powered from the control panel, connecting to the + and
– terminals of K6. RS485 data connects to the A and B terminals, ensuring they are correctly matched at the peripheral device.

No additional resistance should be fitted across the data pair at any point.

Some applications, such as the Pager Interface, take power from the RS485 connector, but do not require the RS485 data connection.

2.3.2 RS232

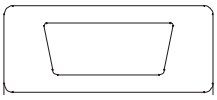


Figure 2.4: RS232

Connector K4 connects the network card RS232 port to peripherals such as Fire Control Panels and Pager systems.

This connection is required to:

1. Control a panel from a remote NCU
2. Connect a computer to a network card for diagnostic purposes or to configure the network card application. Configuration can be performed from any network card on the system
3. Connect a Pager base to the network

The cable used to connect either a computer or a panel to the RS232 port must be a Null Modem cable, which is the same cable used to connect Pertronic analogue addressable control panels to a computer. Connection of peripherals, such as a Pager system, requires a straight-through connector lead.

2.3.3 Network

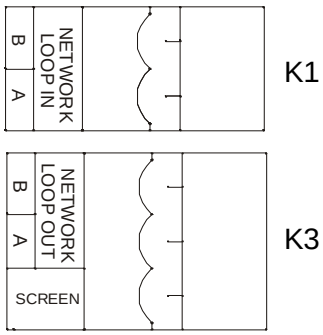


Figure 2.5 : Network

Two connectors K1 and K3 provide high-speed fault tolerant RS485 data connections between the network cards. Each link between one network card and the next is individually monitored for communications failure and Earth faults.

The network cables must be run in the correct sequence - the incoming cable connects to K1 while the outgoing data cable connects to K3 with the terminals marked A and B connecting to similarly marked terminals at the next network card.

If screened cable is used, the screen of the outgoing cable only must connect to the terminal on K3 marked ‘Screen’.

For maximum protection from electrical interference, do not connect the screen to the destination network card, or to Earth any point. The Earth connection is made separately – refer to Section 2.7 ‘Earthing’.

For Earth fault diagnostic purposes, it is advisable to ensure the cable leaving the control panel cabinet is always from a Network Panel card connected directly to that panel, as opposed to a Network NCU card or Network LED Mimic network card. An Earth fault on that network cable will then be identified by that panel.

2.3.4 Flash Programmer

Connector K5, adjacent to the RS232 port, is a 10-way IDC connector, specifically to connect a Pertronic Industries ‘AVR Programmer’ cable, which can be utilised by the “AVR Programmer” or “PonyProg” software to upgrade the firmware. Alternately a 4d Electronics FISP programmer may be used.

This allows the flash program stored on the circuit board to be upgraded. This may be necessary if a new card is added to an existing system and a compatibility issue exists between different versions of firmware. It is recommended that all network cards on the network have the same version of firmware.

NOTE: because there are two versions of the network card, Network Panel and Network NCU, there are also two versions of firmware. It is imperative that the correct version of firmware is used for the card type, as the two card types use identical hardware, with no physical differences between them. The RS485 and Flash Programmer ports use the same internal port, and therefore it is recommended that the RS485 data pair be disconnected during Flash programming to avoid spurious data signals on the RS485 bus.

2.4 Addressing

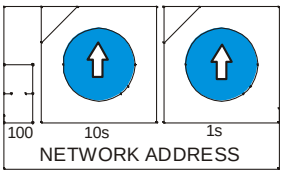


Figure 2.6: Addressing

Each network card requires a unique address. This is set with two rotary dials, SW1 and SW2, in the centre of the card.

The valid Address range is 1 to 99, address 0 being reserved. It is not necessary to keep the addresses consecutive or in any order at all, although for simplicity, it helps to start at Address 1 and be addressed consecutively.

2.5 Indicators

There are eight (8) diagnostic LED indicators, in 4 pairs, on the network cards. Each pair indicates the status of one of the communications ports.

The LED assignments are:

LED1	Receive status for incoming network communications
LED2	Transmit status for incoming network communications
LED3	Receive status for outgoing network communications
LED4	Transmit status for outgoing network communications
LED5	Receive status for the RS232 port communications
LED6	Transmit status for the RS232 port communications
LED7	Receive status for the RS485 or Flash Programmer port communications
LED8	Transmit status for the RS485 or Flash Programmer port communications

2.6 Controls

2.6.1 Reset:

The Reset (SW3) is a momentary pushbutton switch, which provides a ‘soft’ reset to the processor.

2.6.2 Isolate:

The Isolate (SW4) is a toggle switch, which disables communications in both directions between control panels and the network. This allows testing on individual panels to occur without events from the isolated system being passed round the network. It also stops panels being affected by events occurring on the network.

The Isolate switch has no function on **Network NCU** or **Network LED Mimic** cards.

2.7 Earthing

Earthing can be a critical part of the installation, especially where electrical interference may exist from other equipment, such as radio systems, pump equipment etc. Each control panel and network card has its own Earth fault monitoring circuitry, but for this to work effectively, it is necessary to follow the following instructions on Earthing.

There are three Earth points on each network card:

a) Earth Point 1: the screen of the network cable itself – refer to Section 2.3.3 ‘Network’.

b) Earth Point 2: the dedicated Earth terminal, marked NETWORK OUT EARTH (SHIELD), centred between the incoming and outgoing network cable connectors. This terminal should be connected directly to Ground, usually an Earth stud on the gear plate or back box. Failure to make this connection disables the ability to monitor for an Earth connection between the network cable and Ground, making it difficult to diagnose resulting site problems. This connection is necessary even if unscreened network cable is used.

c) Earth Point 3: a dedicated Earth terminal, marked PANEL EARTH, on the lower right corner of the network card circuit board. This connector should be grounded in all cases where the network card is more than 5 metres from the power supply, usually the nearest control panel. If the distance is less than this, the Earth connection will have little benefit but has no detrimental effect and is good practice.

3. NETWORK DISPLAY UNIT (NDU) (OBSOLETE)

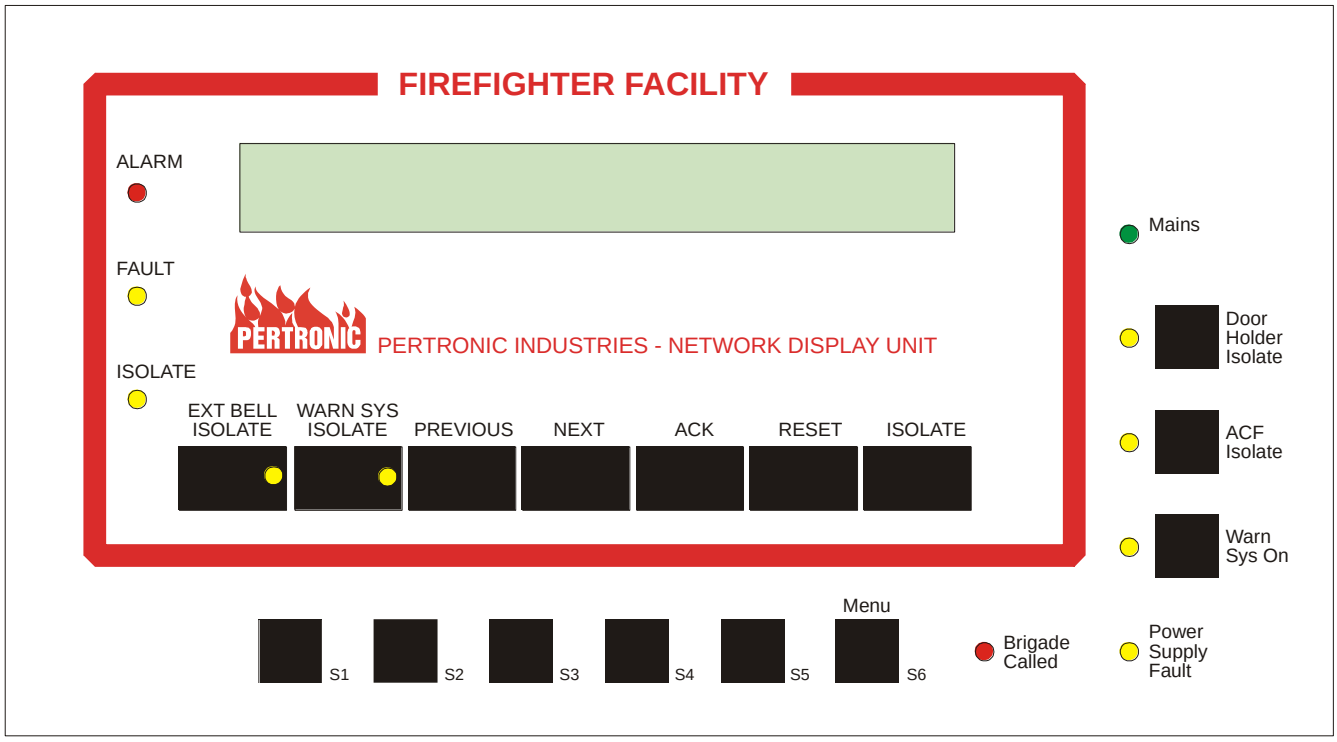


Figure 3.1: Network Display Unit

3.1 Functional Description

The Network Display Unit (NDU) allows an operator to perform limited functions such as:

- isolate and reset active 'Alarm' inputs across the network.
  -an input in 'Normal' cannot be isolated
  -a zone cannot be isolated, however zones may be de-isolated
  -inputs cannot be tested
- set the local date and time (display address 0 only)
- monitor the system status and review current active events.

The display panel contains a two-line by 40-character alpha-numeric LCD display, pushbuttons and LED indicators.

The LCD display itself is within the Fire-Fighter Facility, (enclosed inside a red border), together with the primary push-buttons and LED indicators.

Outside the Fire-fighter Facility are additional engineering controls (push-buttons and LED indicators) for use by fire alarm engineers and the building owner.

When an alarm occurs the NDU overrides the display to view the 'Alarm'. Pushbuttons within the Fire-Fighter area can then be used to Acknowledge, Reset or Isolate the 'Alarm' as required.

3.2 Specification

Dimensions: H x W x D (mm)	200 x 375 x 50		
Power Supply:	Quiescent	6mA @ 27Vdc	- with Backlight and Buzzer OFF
	Alarm	44mA @ 27Vdc	- with Backlight and Buzzer ON
Network Communications:	RS485 Serial Link		
	A	Data 'A' Leg	
	B	Data 'B' Leg	

3.3 Connections

All connectors are on the back of the circuit board.

Connector **K2** makes the power and RS485 data connection to the network card or previous NDU - the connections must be identical.

Connector **K10** makes the power and RS485 connection to the next NDU in the chain, if more than one is installed.

The RS485 address is set by the jumper block **LK3** to **LK6**, in the centre of the board, marked **Address**.

NDUs must be set to consecutive addresses, starting from 0 (no jumpers fitted) up to a maximum of 7 (jumpers 1, 2 and 4 fitted). It does not matter where the addresses are fitted in the RS485 chain, only that they exist from 0 upwards.

Connector **K5** connects an optional external buzzer, for situations where the NDU may be mounted remote from permanent personnel. The buzzer must operate at 27Vdc and the polarity must be followed.

The trim pot marked **VR1** and **Contrast**, adjusts the contrast of the LCD for optimum visibility in the ambient light conditions and viewing angle. If the display is not clear after installation, carefully adjust this pot using a small screwdriver.

Connector **K9** can be used to connect normally-open switches for Warning System and/or Ext Bell Isolate. These switches activate globally, affecting all control panels.

3.4 System Indicators

The Network Display Unit has an LCD Display and eleven (11) LED indicators:

3.4.1 ALARM LED

A red LED indicating:

OFF	the Normal State
Flashing	an unacknowledged ‘Alarm’ event is present in the Fire queue.
ON	one or more ‘Alarm’ events are in the Fire queue and all have been Acknowledged.

3.4.2 FAULT LED

A yellow LED indicating:

OFF	the Normal State
Flashing	one or more ‘Fault’s are present on the system. A ‘Fault’ is a situation that prevents the Fire Alarm System from performing its normal functions. Possible faults include problems with external wiring, defective detectors, component failure, faulty charger or low standby battery voltage.

3.4.3 ISOLATE LED

A yellow LED indicating:

OFF	the Normal State.
ON	one or more devices, (zones, network inputs or logic functions), is isolated. These devices will not activate in the event of an Alarm.

3.4.4 EXT BELL ISOLATE LED

A yellow LED indicating:

OFF	the Normal State
ON	all the External Bell Outputs on all control panels have been isolated by a Network Display (NDU or NCU) ‘External Bell Isolate’ button. These devices will not activate in the event of an Alarm.

3.4.5 WARN SYS ISOLATE LED

A yellow LED indicating:

OFF the Normal State

ON all Warning System Outputs and all loop relays configured as 'Warning System' on all control panels have been isolated by a Network Display (NDU or NCU)'Warning System Isolate' button.

These devices will not activate in the event of an Alarm.

3.4.6 MAINS LED

A green LED indicating:

ON the Normal State with power available to the Network Display (NDU or NCU).

OFF power to the Network Display (NDU or NCU) is absent – the unit is dead.

3.4.7 DOOR HOLDER ISOLATE LED

A yellow LED indicating:

OFF the Normal State

ON all Door Holder Relays and any loop relays configured as 'DOOR' on all control panels have been isolated by a Network Display (NDU or NCU)' Door Holder Isolate' button, and are prevented from energising in the 'Alarm' condition.

3.4.8 ACF ISOLATE LED

A yellow LED indicating:

OFF the Normal State

ON all ACF Relays and any loop relays configured as 'ACF' on all control panels have been isolated by a Network Display (NDU or NCU) 'Isolate ACF' button and are prevented from energising in the 'Alarm' condition.

3.4.9 WARN SYS ON LED

A yellow LED indicating:

OFF the Normal State

ON all Warning Systems and External Bells on all control panels have been activated by a Network Display (NDU or NCU) 'Warn Sys On' button.
This function cannot be over-ridden by another function except the Ext Bell and Warning System Isolate.

3.4.10 POWER SUPPLY FAULT LED

A yellow LED indicating:

OFF the Normal State, indicating that all the control panel standby batteries and their associated chargers are normal.

ON a Power Supply fault exists on one or more of the control panels in the network.

3.4.11 BRIGADE CALLED LED

A red LED indicating:

OFF the Normal State

ON a 'Fire' signal has been sent to the Brigade-calling device by at least one of the control panels on the network.

3.5 Fire-Fighter Facility

The wide red border as shown in Figure 3.1 encloses the Fire-Fighter Facility (FF).

Note that any button pressed after an event silences the local buzzer.

The seven push buttons within the FF have the following functions:

3.5.1 EXT BELL ISOLATE Button

The EXT BELL ISOLATE push-button isolates the External Bells across the entire network.

When the push-button LED is ON, the External Bells are disabled. External Bells are turned OFF if already active. -

When the LED is OFF, the External Bells are enabled.

This switch has a toggle function as alternate presses of EXT BELL ISOLATE will enable or disable the External Bells.

3.5.2 WARN SYS ISOLATE Button

The WARN SYS ISOLATE push-button disables the Warning System relays across the entire network. When the push-button LED is ON, the WARNING SYSTEM relays are isolated. WARNING SYSTEM relays are turned OFF, if already active.

This switch has a toggle function as alternate presses of WARN SYS ISOLATE will correspondingly disable or enable the use of the WARNING SYSTEM.

In a similar manner, WARN SYS ISOLATE also disables Loop Relays configured as 'Warning System' relays.

3.5.3 PREVIOUS Button

The PREVIOUS push-button scrolls current queue events on the LCD display. Each time the button is pressed, the next-oldest event is displayed until the oldest is reached, at which point the display scrolls and the most recent event is displayed again.

3.5.4 NEXT Button

The NEXT push-button operates in the same way as PREVIOUS, except that the next most-recent event is displayed each time the button is pressed until the latest event is reached, at which point the display scrolls and the most recent event is displayed again.

3.5.5 ACK Button

The ACK push-button acknowledges the alarm currently displayed on the LCD.

If there is only one un-acknowledged alarm on the system, the local buzzer is silenced and the ALARM LED stops flashing and remains ON. If there are more un-acknowledged alarms, the ALARM LED remains flashing.

The ACK button also confirms device Isolation and Reset in the Event queue.

3.5.6 RESET Button

The RESET push-button resets the device currently displayed in an 'Alarm' state.

The NDU will request that ACK be pressed to confirm the reset operation. If this is not done within 10 seconds, or if any other key is pressed, the reset function will be cancelled.

The Reset button can also be used to reset devices or zones in the 'Isolate' queue.

3.5.7 ISOLATE Button

The ISOLATE push-button performs a range of functions dependent on the event currently displayed.

Alarm Queue	isolating a device in the 'Alarm' queue causes that device to reset from the 'Alarm' state and be removed from the 'Alarm' queue and placed in the 'Isolate' queue.
Isolate Queue	the Isolate button de-isolates the device currently displayed in the 'Isolate' queue. If the device is in 'Alarm' a warning will be given. The device or zone is automatically reset on de-isolation.

Confirmation by activation of the ACK pushbutton is prompted for and required within 10 seconds of an Isolate operation else the operation will be cancelled.

3.6 Engineering / Operator Controls

The controls outside the Fire-Fighter Facility, as shown in Figure 3.1 are for use by service and maintenance personnel or by the building owner.

The push buttons along the bottom of the front panel are multi-function buttons, used to navigate the menu system and also for edit / control options from within the menus.

3.6.1 S1 to S5

Used as general-purpose function buttons, as displayed within the relevant menu.

3.6.2 S6 - MENU

Used as general-purpose function button and to Enter and Exit the menu.

3.6.3 DOOR HOLDER ISOLATE

The DOOR HOLDER ISOLATE push-button isolates the Door Holder relays and all Loop Relays configured as ‘Door Holder’ relays across the entire network. When the push-button LED is ON, the Door Holder relays are isolated.

When the LED is OFF, the Door Holder relays are enabled.

This switch has a toggle function as alternate presses of ‘Door Holder Isolate’ will correspondingly enable or isolate the Door Holder relays.

All Door Holder relays are de-energised when the ‘Door Holder Isolate’ function is active.

3.6.4 ACF ISOLATE

The ACF ISOLATE push-button isolates the ACF relays and all Loop Relays configured as ‘ACF’ across the entire network. When the associated LED is ON, the ACF relays are isolated.

When the LED is OFF, the ACF relays are enabled.

This switch has a toggle function as alternate presses of ACF Isolate will correspondingly enable or isolate the ACF relays.

The ACF relays are de-energised when the ‘ACF Isolate’ function is active.

3.6.5 WARN SYS ON

When the WARN SYS ON function is active (LED ON), the WARNING SYSTEM and EXTERNAL BELLs are activated.

This function cannot be over-ridden by any other operation except ‘EXT BELL ISOLATE’ and/or ‘WARN SYS ISOLATE’.

The Isolate function over-rides the ‘Warn Sys On’ command.

This switch has a toggle function as alternate presses of ‘Warn Sys On’ will correspondingly activate or de-activate the External Bell and Warning System.

3.7 Menu Options

Press the **Menu** (**S6**) pushbutton to open a menu with the following options:

```
Select Function (S1, S2, S3 OR S6)
Clk   Flt    Hist                  Exit
```

Select the **Menu** (**S6**) button to EXIT from the menu to the normal display.

3.7.1 Clock

Select the Clk option using the **S1** pushbutton, to change the time and date settings of the NDU display. Note that an NDU display set to RS485 address 0 will update all other Date and Time displays on the network. NDU displays set to address 1 or higher cannot be used to set the Date and Time.

```
Set Time/Date
_13:53:11 07/03/08
```

The cursor will initially sit under the Hours figure (24 hour format).

The pushbuttons Previous and Next can be used to move the cursor backward or forward to each field in turn, and the **S1** and **S2** function keys Decrement or Increment the selected figure.

When all figures are correct, press the **Menu** (**S6**) button to return to the menu.

3.7.2 Fault

Select the Flt option using the **S2** pushbutton, to set the local Fault buzzer option.

```
Fault Buzzer is:  Enabled
Disable                          Exit
```

Press the **S1** pushbutton to toggle the Fault buzzer to Enable or Disable.

When set to disabled, the local NDU buzzer will not sound for any 'Fault' conditions, but will activate on 'Alarm'.

This setting is held in volatile memory and will be lost if power is removed from the network card or the network card firmware is upgraded.

Press the **Menu** (**S6**) pushbutton to exit this screen.

3.7.3 History

Select the Hist option using the **S3** pushbutton, to enter the network history event log screen.

```
0000: Panel Door Open
Panel 001        At: 13:18:38 23/08/08
```

The pushbuttons Previous and Next can be used to scroll through the logged events. **S2** can be used to advance 10 subsequently logged events, **S3** advances through 10 previously logged events. Pushbutton **S1** displays the most recent event (event 0). **S6** can be used to exit the history screen and return to the menu.

Top line of display has the history log event number followed by the event description. Bottom line of the display has event details followed by the date and time the event was logged.

Network History Log function only works for firmware versions v3.0 (v2.20 PCB) and greater.

3.8 Mini Mimics

LCD Mini-Mimics (F100AMM-3 or MINIMIMIC) having a firmware version of 2.28 or later may be connected to the network in place of any full NDU display. The maximum number of Network Display Units (NDU) and Mini-Mimics which can be connected to each 'NDU' network card is eight (8). LCD Mini-Mimics may use any of the addresses from 0 to 7. LCD Mini-Mimics with firmware versions prior to 2.28 are restricted to addresses 1-7 only and must have a full NDU display at address 0. Refer to the Mini Mimic Technical Manual for details of correct configuration.

As a Network LCD Mini-Mimic set to address 0 only includes the 'Buzzer Silence' and 'Next' buttons, it is only possible to silence the buzzer and scroll to the next event in the highest priority network queue. It is not possible to change the network queue displayed.

4. NETWORK CONTROL UNIT (NCU)

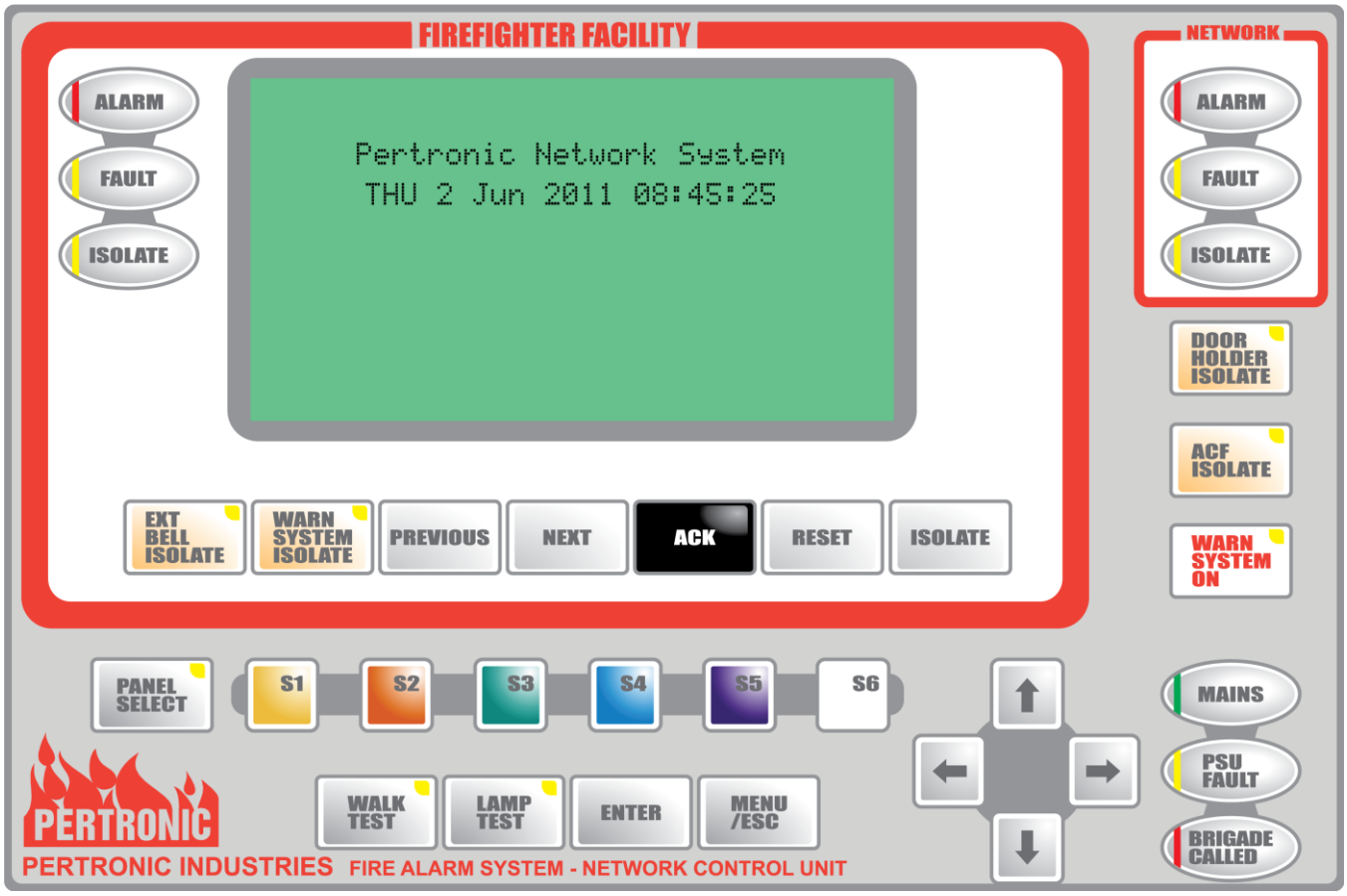


Figure 4.1: Network Control Unit

4.1 FUNCTIONAL DESCRIPTION

The Network Control Unit (NCU) allows an operator to perform routine functions:

- Isolate, test and reset 'Alarm' inputs and zones across the entire network
- Set the local date and time
- Monitor the system status and review logged events.
- Remotely connect to network panels in order to operate individual panel functions.

The display panel contains a graphics LCD display (240 x 128 pixel), pushbuttons and indicators.

The LCD display itself is within the Fire-Fighter Facility, (enclosed inside a red border), together with primary push-buttons and LED indicators.

Outside the Fire-fighter Facility are additional engineering controls (push-buttons and LED indicators) for use by fire alarm engineers and the building owner.

When an alarm occurs the NCU overrides the display to view the 'Alarm'. Pushbuttons within the Fire-Fighter area can then be used to Acknowledge, Reset or Isolate the 'Alarm' as required.

4.2 SPECIFICATION

Dimensions: H x W x D (mm)	232 x 375 x 50		
Power Supply:	Quiescent	20mA @ 27Vdc	- with Backlight and Buzzer OFF
	Alarm	40mA @ 27Vdc	- with Backlight and Buzzer ON
Network Communications:	RS485 Serial Link		
	A	Data 'A' Leg	
	B	Data 'B' Leg	

4.3 CONNECTION AND MOUNTING

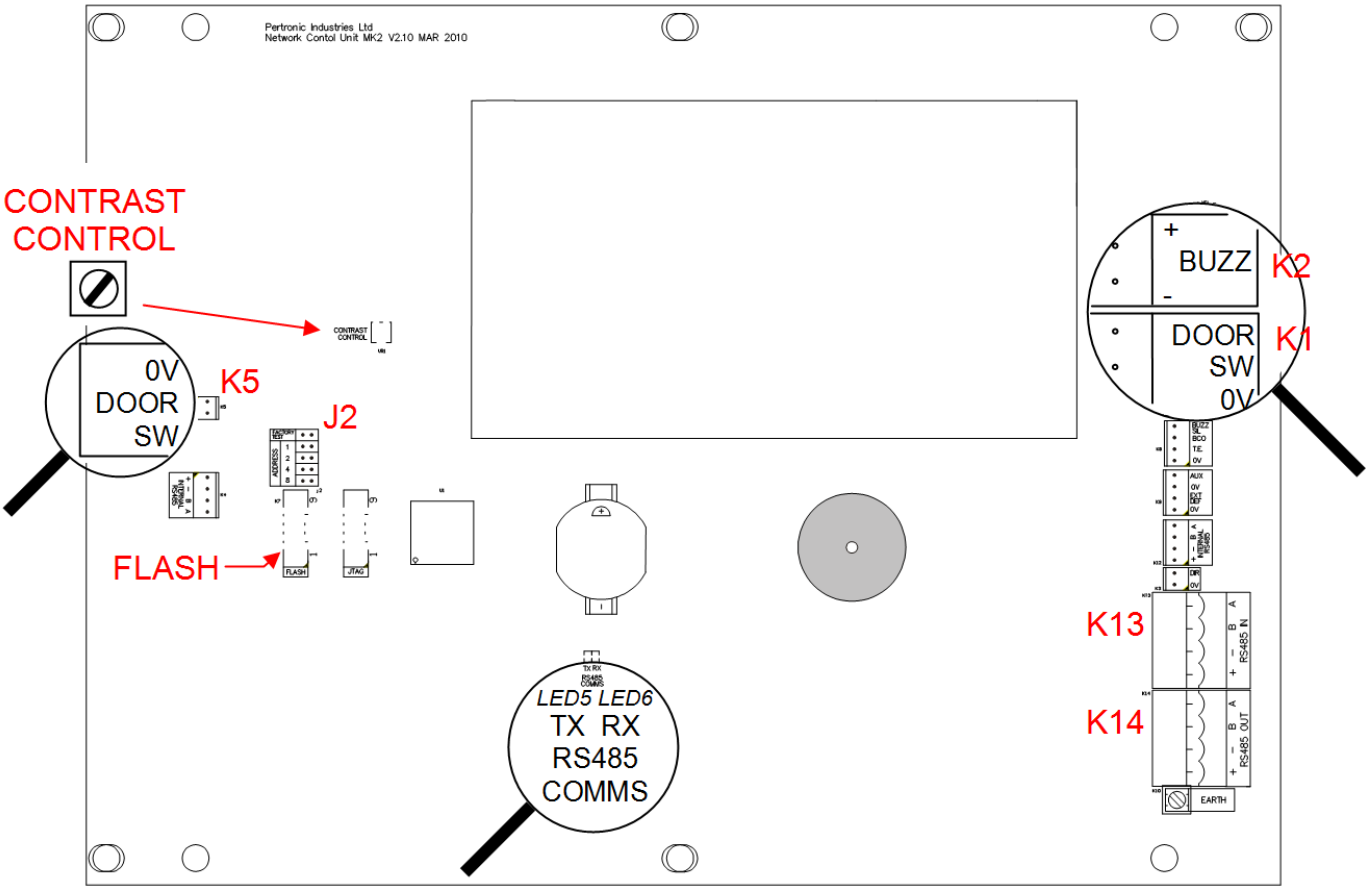


Figure 4.2: Network Control Unit—rear view

Power and RS485 Data connection from the network card are made to connector **K13**, marked **RS485 IN**, observing each cable is similarly marked at each end, i.e.: A to A, + to + etc.

The door switch pair of cables connects to **K1** or **K5**, marked **DOOR SW** and **0V**. The orientation is not important unless the switch is also connected to other connectors, such as a control panel.

Where the NCU is mounted as a stand-alone unit, the cabinet will have its own door switch. However, if the NCU is mounted inside a control panel, in place of the panel's own display and keyboard, the panel's door switch should be connected to connector **K1** or **K5** instead of the panel's door switch input, which should be bridged out.

An external buzzer may be connected to connector **K2**, if required to supplement the on-board buzzer. This should be a 12Vdc buzzer.

The graphics LCD contrast may be adjusted using a small screwdriver. Turn trim pot **VR1**, marked **CONTRAST CONTROL**, to obtain optimum contrast in the installed ambient light conditions and viewing angle.

The Pertronic Industries Ltd AVR programming tool may be used to upgrade the firmware of the Universal KB Display Driver board if required, using connector **K5**, marked **FLASH**.

The address of the NCU (as distinct from the address of the network card) is set by fitting a jumper on connector **J2** marked **ADDRESS**, to short out the two pins marked 1. Only one NCU can be connected to a network card and it should always be set to Address 1.

An on-board battery powers the real time clock on the Universal KB Display Driver board. If the date and time are not maintained after a power down, the battery may need replacement. Replace with a 3V type CR2032 or suitable alternative battery, and reset the date and time.

4.3.1 SYSTEM INDICATORS

The Network Control Unit has an LCD Display and seventeen (17) visible LED indicators.

Apart from the Panel Select LED and the three LEDs in the separate section marked Network, all other indicators reflect the status of the global network when in normal supervisory mode.

However, if the Panel Select pushbutton is used to monitor an individual control panel, the LED indicators mimic those of the target control panel only.

The three (3) Network LEDs (Alarm, Fault and Isolate), show the status of the global network at all times.

4.3.1.1 ALARM LED

A red LED indicating:

OFF	the Normal State
Flashing	an unacknowledged ‘Alarm’ event is present in the ‘Fire’ queue.
ON	one or more ‘Alarm’ events are in the ‘Fire’ queue and all have been Acknowledged.

4.3.1.2 FAULT LED

A yellow LED indicating:

OFF	the Normal State
Flashing	one or more Fault’s are present on the system. A ‘Fault’ is a situation that prevents the Fire Alarm System from performing its normal functions. Possible faults include problems with external wiring, defective detectors, component failure, faulty charger or low standby battery voltage.

4.3.1.3 ISOLATE LED

A yellow LED indicating:

OFF	the Normal State.
ON	one or more devices (zones, network inputs or logic functions), is isolated.

4.3.1.4 EXT BELL ISOLATE LED

A yellow LED indicating:

OFF	the Normal State
ON	all the External Bell Outputs on all control panels have been isolated by a Network Display (NDU or NCU) External Bell Isolate button. External Bells will not activate in the event of an Alarm.

4.3.1.5 WARN SYS ISOLATE LED

A yellow LED indicating:

OFF	the Normal State
ON	all the Warning System Outputs and all Loop Relays configured as ‘Warning System’ on all control panels have been isolated by a Network Display (NDU or NCU) Warning System Isolate button. Warning Systems will not activate in the event of an Alarm.

4.3.1.6 DOOR HOLDER ISOLATE LED

A yellow LED indicating:

OFF	the Normal State
ON	all the Door Holder Relays and any Loop Relays configured as ‘DOOR’ on all control panels have been isolated by a Network Display (NDU or NCU) Door Holder Isolate button, and are prevented from energising in the ‘Alarm’ condition.

4.3.1.7 ACF ISOLATE LED

A yellow LED indicating:

OFF	the Normal State
ON	indicates that all the ACF Relays and any Loop Relays configured as 'ACF' on all control panels have been isolated by a Network Display (NDU or NCU) Isolate ACF button and are prevented from energising in the 'Alarm' condition.

4.3.1.8 WARN SYS ON LED

A yellow LED indicating:

OFF	the Normal State
ON	indicates that all Warning Systems and External Bells on all control panels have been activated by a Network Display (NDU or NCU) Warn Sys On button. This function cannot be over-ridden by any other functions except the Ext Bell and/or Warning System Isolate.

4.3.1.9 MAINS LED

A green LED indicating:

OFF	the NCU has no power and is not functioning
ON	the NCU has power and is functioning

4.3.1.10 PSU FAULT LED

A yellow LED indicating:

OFF	the Normal State, indicating that all of the control panels' standby batteries and their associated chargers are Normal.
ON	indicates that there is a Power Supply fault with one or more of the control panels in the network.

4.3.1.11 BRIGADE CALLED LED

A red LED indicating:

OFF	the Normal State
ON	indicates that a fire signal has been sent to the Brigade-calling device by at least one of the control panels on the network.

4.3.1.12 NETWORK – ALARM LED

A red LED indicating:

OFF	the Normal State
Flashing	an unacknowledged 'Alarm' event is present in the Fire queue.
ON	One or more 'Alarm' events are in the Fire queue and all have been Acknowledged.

4.3.1.13 NETWORK – FAULT LED

A yellow LED indicating:

OFF	the Normal State
Flashing	one or more Faults are present on the system. A 'Fault' is a situation that prevents the Fire Alarm System from performing its normal functions. Possible faults include problems with external wiring, defective detectors, component failure, faulty charger or low standby battery voltage.

4.3.1.14 NETWORK – ISOLATE LED

A yellow LED indicating:

OFF	the Normal State
ON	One or more devices on the network (zones, network inputs or logic functions), is isolated.

4.4 FIRE-FIGHTER FACILITY

The wide border as shown in Figure 3.3: Network Display Unit encloses the Fire-Fighter Facility (FFF).

Note that any button pressed after an event silences the buzzer globally (all panels and supervisory displays).

In the F120 panel's Takeover Mode, a key press generates two responses:

******* immediately across the top of the screen, indicating the Display recognises the key press,

and then an audible beep, indicating the Panel recognises the key press.

In the F100 panel's Takeover Mode AND in Global Mode, the same responses are generated; however, both responses are generated by the Display—and NOT by the Panel.

The seven (7) push buttons within the FFF have the following functions:

4.4.1 EXT BELL ISOLATE BUTTON

The EXT BELL ISOLATE push-button isolates the External Bells across the entire network.

When the push-button LED is ON, the External Bells are disabled. External Bells are turned OFF if already active.

When the LED is OFF, the External Bells are enabled.

This switch has a toggle function as alternate presses of EXT BELL ISOLATE will enable or disable the External Bells.

This key may be selectively enabled and disabled from the relevant “factory test” menu (See section 4.6.6). Its operation will be dependent on what is set there.

4.4.2 WARN SYS ISOLATE BUTTON

The WARN SYS ISOLATE push-button disables the Warning System relays across the entire network. When the push-button LED is ON, the WARNING SYSTEM relays are isolated. WARNING SYSTEM relays are turned OFF, if already active.

This switch has a toggle function as alternate presses of WARN SYS ISOLATE will correspondingly disable or enable the use of the WARNING SYSTEM.

In a similar manner, WARN SYS ISOLATE also disables Loop Relays configured as ‘Warning System’ relays.

This key may be selectively enabled and disabled from the relevant “factory test” menu (See section 4.6.6). Its operation will be dependent on what is set there.

4.4.3 PREVIOUS BUTTON

The PREVIOUS push-button scrolls current queue events on the LCD display. Each time the button is pressed, the next-oldest event is displayed until the oldest is reached, at which point the display scrolls and the most recent event is displayed again.

4.4.4 NEXT BUTTON

The NEXT push-button operates in the same way as PREVIOUS, except that the next most-recent event is displayed each time the button is pressed until the latest event is reached, at which point the display scrolls and the most recent event is displayed again.

4.4.5 ACK BUTTON

The ACK push-button acknowledges the alarm currently displayed on the LCD.

If there is only one un-acknowledged alarm on the system, the local buzzer is silenced and the ALARM LED stops flashing and remains ON. If there are more un-acknowledged alarms, the ALARM LED remains flashing.

The ACK button also confirms device Isolation and Reset in the event queues.

4.4.6 RESET BUTTON

The RESET push-button resets the device currently displayed in an ‘Alarm’ state.

The NCU will request that ACK be pressed to confirm the reset operation. If this is not done within 10 seconds, or if any other key is pressed, the reset function will be cancelled.

4.4.7 ISOLATE BUTTON

The ISOLATE push-button performs a range of functions dependent on the event currently displayed.

Alarm Queue isolating a device in the ‘Alarm’ queue causes that device to reset from the ‘Alarm’ state and be removed from the ‘Alarm’ queue. The device is then placed in the ‘Isolate’ queue.

Isolate Queue the Isolate button de-isolates the device currently displayed in the ‘Isolate’ queue.

If the device is in ‘Alarm’ a warning will be given. The device or zone is automatically reset on de-isolation.

Confirmation by activation of the ACK pushbutton is prompted for and required within 10 seconds of an Isolate operation else the operation will be cancelled.

4.5 ENGINEERING / OPERATOR CONTROLS

The controls outside the Fire-Fighter Facility are for use by service and maintenance personnel or by the building owner. These are shown in Figure 4.4: Network Control Unit.

The push buttons along the bottom of the front panel are multi-function buttons, used to navigate the menu system and also for edit / control options from within the menus.

In the F120 panel's Takeover Mode, a key press generates two responses:

1. ******* immediately across the top of the screen, indicating the Display recognises the key press,
2. and then an audible beep, indicating the Panel recognises the key press.

In the F100 panel's Takeover Mode AND in Global Mode, the same responses are generated; however, both responses are generated by the Display—and NOT by the Panel.

4.5.1 S1 TO S5

Used as general-purpose function buttons, these are displayed within the relevant menus.

4.5.2 S6 - MENU

Used as general purpose function button and to enter and exit the menu.

4.5.3 DOOR HOLDER ISOLATE

The DOOR HOLDER ISOLATE push-button isolates the door holder relays and all Loop Relays configured as ‘Door Holder’ relays across the entire network. When the push-button LED is ON, the door holder relays are isolated.

When the LED is OFF, the door holder relays are enabled.

This switch has a toggle function as alternate presses of ‘Door Holder Isolate’ will correspondingly enable or isolate the door holder relays.

All door holder relays are de-energised when the ‘Door Holder Isolate’ function is active.

This key may be selectively enabled and disabled from the relevant “factory test” menu (See section 4.6.6). Its operation will be dependent on what is set there.

4.5.4 ACF ISOLATE

The ACF ISOLATE push-button isolates the ACF relays and all Loop Relays configured as ‘ACF’ across the entire network. When the associated LED is ON, the relays are isolated.

When the LED is OFF, the relays are enabled.

This switch has a toggle function as alternate presses of ACF Isolate will correspondingly enable or isolate the relays.

The auxiliary relays are de-energised when the ‘ACF Isolate’ function is active.

This key may be selectively enabled and disabled from the relevant “factory test” menu (See section 4. 6.6). Its operation will be dependent on what is set there.

4.5.5 WARN SYS ON

When the WARN SYS ON function is active (LED ON), the WARNING SYSTEM and EXTERNAL BELLs are activated.

This function cannot be over-ridden by any other operation except ‘EXT BELL ISOLATE’ and/or ‘WARN SYS ISOLATE’.

The Isolate function over-rides the ‘Warn Sys On’ command.

This has a toggle function as alternate presses of ‘Warn Sys On’ will correspondingly activate or de-activate the External Bell and Warning System.

This key may be selectively enabled and disabled from the relevant “factory test” menu (See section 4.6.6). Its operation will be dependent on what is set there.

4.5.6 WALK TEST

The WALK TEST button has no function while in ‘Normal’ mode or when used in conjunction with an F100A control panel.

If an F120A control panel has been selected using the Panel Select function, the WALK TEST button mimics that of the panel itself and can be used to put the selected panel into Walk Test.

4.5.7 LAMP TEST

The LAMP TEST button can be used at any time to instigate a Lamp Test function.

In ‘Normal’ mode, the test operates all NCU Display LED’s in turn, and performs a pattern test on the graphics LCD.

In ‘Panel Monitor’ mode with an F120 selected, the indicator LED’s on the NCU mimic those of the selected control panel.

Notes:

- Some LED’s on the NCU do not exist on the F120 and will therefore not be activated during this test.
- LED’s activated on the NCU do not imply that the LED’s on the panel are actually functioning, only that the panel is attempting to turn them on.

The LAMP TEST button has a toggle function and alternate presses correspondingly activate or de-activate the test sequence.

The F100A control panel cannot perform a Lamp Test with this button but can be tested from the F100A Panel menu.

4.5.8 ENTER

The ENTER pushbutton has two main functions:

- In ‘Normal’ mode, it toggles between active event queues, for example, to switch between ‘Alarms’ and ‘Faults’.
- In ‘Panel Monitor’ mode, the pushbutton mimics that of the control panel and is used to accept or select options.

4.6 MENU

In ‘Panel Monitor’ mode, the **MENU** and **S6** buttons have the same function, mimicking the **MENU** button of the control panel. In ‘Normal’ mode, the **MENU** pushbutton opens the NCU menu:

```
Select Function (S1, S2, S3 OR S6)




Clk   Flt    Hist                   Exit
S1     S2      S3      S4      S5     S6
```

Select the S6 or MENU button to EXIT from the menu to the normal display.

4.6.1 CLOCK

Select the Clk option using the **S1** pushbutton to change the Date and Time settings of the NCU display.

```
            23:59:21  30/06/08

          Current display time:
            23:59:37  30/06/08


[Down]  [Up]  [Left]  [Right]  [Save]  [Esc]
[Reset]
    S1    S2    S3     S4     S5     S6
```

The cursor will initially blink over the first character of the Hours figure (24-Hour format).

Press **S1** or **Down** to decrement the currently selected digit, which is underscored and flashing on the LCD screen—shown shaded grey above. Press **S2** or **Up** to increment the selected number.

Press **S4** or **Right** to move and select the next digit; press **S3** or **Left** to do the same for the previous digit.

Press **S5** or **Save** to retain the clock setting and exit the menu.

Press **S6** or **Menu** to exit the menu without saving.

If you want to reset the “edit time”—top line of display—to conform to the Current display time, press the **Reset** button.

If the NCU is configured as Network Clock, the Network time will be updated shortly after pressing **Enter** or **S5**.

4.6.2 FAULT

Select the Flt option using push-button **S2**, to set the local Fault buzzer option.

```
Fault Buzzer is:  Enabled




Disable                           Exit
S1     S2     S3       S4     S5      S6
```

Press push-button **S1** to toggle the Fault buzzer to either Disabled or Enabled.

When set to disabled, the local NCU buzzer will not sound for any ‘Fault’ conditions, but will activate on ‘Alarm’. This setting is held in volatile memory and will be lost if power is removed from the network card or the network card firmware is upgraded.

Press the Menu push-button (**S6**) to exit this screen.

4.6.3 NETWORK HISTORY LOG

```
              Network History
0000: Panel Door Open
Panel 051
Desc:  Panel Descriptor
Logged at 13:17:56, 23/08/08

Top    -10    +10
S1     S2     S3      S4     S5      S6
```

Select the Hist option using the **S3** pushbutton to enter the network history log screen. Press push-button **S1** to display the most recent event. Push-buttons next and previous can be used to scroll through logged events. **S2** can be used to advance 10 subsequently logged events, **S3** advances through 10 previously logged events. **S6** can be used to exit the history screen and return to the menu.

Second line of display has the history log event number followed by the event description. The third line of the display has event details. The fourth line of the display has the panel descriptor of the panel the event is from. The fifth line of the display has the date and time the event was logged.

Network History Log function only works for firmware versions v3.0 and greater.

4.6.4 PANEL SELECT

The PANEL SELECT pushbutton is used to enter Panel Monitor mode. The user may select an individual control panel from a list of those present on the network. The default panel to connect to may be configured in the Network Utility. Only one panel can be monitored on a network at any one time, even if more than one NCU is present. Attempts to make a second connection will result in a Connection Error message being displayed on the LCD.

The example below shows that the control panel in the Security Suite of the building, which has a network card set to address 002, is the panel currently selected. Press the S5 function button to select that panel, S6 or MENU will return to the normal screen. **S1** will select the previous configured panel; **S2** will select the next configured panel.

```
Select Panel       (S5/Enter to Connect)
Network Address:002
Panel: Panel 2



Prev  Next                   Connect  Exit
S1     S2     S3       S4     S5      S6
```

Once a Panel has been selected, the upper section of the LCD screen and NCU indicator LEDs will briefly indicate that the connection is being made to the selected panel, and will then mimic that of the control panel. The lower section of the screen will now indicate the network address and descriptor of the connected control panel. The user can perform all display and keyboard functions as though they were at the remote control panel, e.g.: check the historic log to ascertain the sequence of events occurring at a specific time.

While the remote panel remains connected, the NCU indicator LED '**PANEL SELECT**' will be ON, indicating that the display is not representative of the entire network.

If an Alarm occurs anywhere on the network whilst a remote panel is connected, the connection to the control panel will be terminated so that the alarm details can be quickly observed. This will occur even if the connected control panel is the one in alarm.

To exit from this mode, press **PANEL SELECT** again to return to the panel selection screen, and then **S6** or **MENU** to return back to the normal screen.

Note: Closing the door to the NCU automatically disconnects the remote connection, and the display will revert to normal mode.

4.6.5 CURSOR KEYS

The cursor keys arrowed LEFT and RIGHT navigate between fields or characters when connected to an F120 control panel, while the UP and DOWN keys scroll the selected fields up or down. This provides a means to scroll through the available panel list to make a remote connection, or adjust the date and time values.

The cursor keys have no function when connected to a remote F100A control panel, because the F100A does not support cursor control keys.

4.6.6 DISABLE KEYS MENU

It may be desirable to disable certain keys from having a global effect. The following keys can be disabled:

- **EXT BELL ISOLATE**
- **WARN SYS ISOLATE**
- **DOOR HOLDER ISOLATE**
- **ACF ISOLATE**
- **WARN SYS ON**

Note: Disablement of a key is done separately for each NCU. So, to disable the keys for an entire site, the disablement procedure must be done for every NCU used at the site.

To access the Disable Keys menu, follow these steps:

1. Put a jumper across the Factory Test terminals on the rear of the display, forcing a hard reset.
2. Enter Country—S2.
3. Select Keyboard—S2.
4. Select Key Disable—S5.
5. Press each key as indicated to cycle through the options available:
    a. Globally Enabled, meaning the key is always active—the default setting.
    b. Locally Enabled, meaning the key only produces a response when the ‘PANEL SELECT’ LED is lit. Note: This also applies when selecting a panel to take over, and not just in Takeover mode itself.
    c. Always Disabled, meaning the key produces no effect.

6. Press **MENU** or **S6** to exit.

7. Remove the jumper across the Factory Text terminals to force a hard reset.

4.7 MAKING JUMPER CHANGES

The NCU will automatically recognize any changes to configuration jumpers. Configuration jumpers are the following:

- Address jumpers (1–8)
- Factory test mode

The NCU automatically restarts itself after a configuration jumper change occurs. Manually unplugging power cables is not necessary to force a restart. However, note that the restarting process itself will likely generate temporary network errors as the NCU goes off line and then on again.

4.8 SILENCING THE LOCAL BUZZER

The NCU includes an internal buzzer and a connection for an optional external buzzer. If sounding, buzzer(s) can be silenced by pressing any enabled key. Keys which have been explicitly disabled, as detailed in section 4.5, will not silence the buzzer.

4.9 USING THE NCU AS A PANEL DISPLAY

If configuring a Panel without a panel display, an NCU may be attached directly to show the locally generated display content. The NCU will automatically detect the Panel type, and it does not require configuration beyond the standard address jumpers. However, such a deployment of the NCU should not be used for a permanent installation. Instead, a panel-specific display intended for that purpose should be utilized.

An NCU can only be connected directly to one Panel at a time. The method used to configure the NCU depends on whether the Panel is an F100 or an F120.

Connecting to an F100 Panel

Under this arrangement, address 0 on an F100 display maps to address 1 on an NCU. That is, all addresses on the NCU should be incremented up by one from the panel address. Thus, in order to connect an NCU to an F100 as a mimic with address 2, you should set the address jumper to 3.

Connecting to an F120 Panel

User this arrangement, all address mappings remain the same address number between panel and NCU. For example, address 1 on an F120 display maps to address 1 on an NCU display.

It should be noted that because the NCU does not support TTL communications, address 0 (which would otherwise be physically connected via 'F120 TTL KEYBOARD' --> 'F120 KEYBOARD') is unavailable. The display should be configured as address 1 upwards, and connected via the Internal RS485 connector on the Panel.

Note that the NCU cannot use the TTL connection on the F120 Panel. Instead, the display should be configured as address 1 or higher and be connected via the internal RS485 connector on the Panel.

5. NETWORK LED DISPLAY OPERATION

The Installation Notes for the LED Display Unit (LDU) and RS485 LED Address Controller (LAC) cover all the functions available and therefore are not included in this manual. However, when used as a Network LED Mimic, there are some differences, which should be noted.

Power and data cables connect the LDU or LAC to the associated **Network Panel** card, not to the Fire Control Panel.

Up to a maximum of 8 network mimics can be driven by a single network card - they should always be set to addresses from 01 to 08 for flashing LEDs, or 11 to 18 for non-flashing LEDs.

All functional switched inputs are global, and affect all network connected panels and displays.
No network mapping is required for these to operate across the network.

All switched outputs and the System LEDs are global, affected by events anywhere on the network.
No mapping is required to activate these outputs.

Events occurring on the network panels can be mapped to a maximum of 128 LEDs (including the existing 8 on the LDU). Typically these will be individual zone alarms, and thus if more than 128 zones are to be indicated, two or more mimics will be required, each with its own network card.

6. PAGER OPERATION

Three Pager systems are currently supported:

1. AUSTCO
2. AUSTCO-APR
3. KEW.

The Pager protocols utilised by the Network cards differs from those used by the GPIB boards, so care must be taken before replacing a GPIB board with a Network Pager card. Refer to Pertronic Tech Support for assistance!

The Pager Base system connects to the Network NCU card directly via the RS232 port, using a cable with the following minimum connections.

Pin Use	Pager (D9)	Pager (D25)	Cable D9/D25 to (D9F)	Network Card (D9F)
Ground	5	7		5
Transmit Data: TxD	3	2		3
Receive Data: RxD	2	3		2

Table 6.1**: Pager Interface**

The network card is configured with the relevant operating parameters for the system, via the Network Configuration Utility. Refer to Network Configuration Program for details. Note that from firmware version 2.0 it is possible to connect a pager base to an existing NCU or NDU network card, once correctly configured.

From network firmware version 2.02 / 3.02 it is possible to send a pager test message at up to 4 pre-configured times a day, configured through the Network Configuration Program version 2.09.03 and above. The device descriptor for the test message is hard coded to “Test Only”.

Refer to the operating manual for the Pager Base for details of the RS232 Cable and for configuring the Pager Base itself.

6.1 Pertronic Network Pager Interface - AUSTCO APR

The AUSTCO APR protocol is used by the Pertronic Network Supervisory card when configured as an AUSTCO_APR Pager, to drive a Pager base. **APR** stands for **A**ustco **P**aging **R**equest.

The communications parameters are:

Type: RS232
Baud Rate: Configurable - 300, 600, 1200, 2400, 4800, 9600, or 19200 (default = 9600)
Data Bits: 8
Stop Bits: 1
Parity: None
Handshaking: None

The default structure of the message is:

<TAB> Pager address/Address type/Descriptor<CR>

An optional **Panel** field can additionally be configured to give the message format:

<TAB> Pager address/Address type/Panel<SP>Descriptor<CR>

The header and trailer are always a tab and carriage return character respectively, which will not print. The field separator is always a “/” slash except where the Panel field is used, in which case a single <space> character is inserted between the Panel and Descriptor fields.

6.2 Pager Address: 1 to 3 characters

A pager address ID (cap code) is a three digit number sent to the Pager base using the indirect method. The letter U precedes a 3 digit unit number NNN ranging from 000 to 999 (U900 in the example below). The software in the Pager base refers to a table that relates unit numbers to cap codes. This table can be edited on site in the Pager base.

Only one Pager ID is supported by the network and is configured in the Pertronic Network Utility.

6.2.1 Address Type: 1 character

Each pager address ID (cap code) has attached an address type of 4. This address type is used by the pager base to trigger a specific beeping tone/sequence in the receiving pager. The paging base software contains an explicit relationship to translate this code to the correct priority message. This relationship can be edited on site in the pager base.

6.2.2 Panel: 1 to 4 optional characters.

Each Fire Control Panel has a dedicated network interface card, which is configured with its own descriptor. If the descriptor is configured (within the Pertronic Network Utility program) to end with **\xxxx** where xxxx can be up to four alphanumeric characters, then the Panel field will be inserted using these characters, followed by a space.

6.2.3 Descriptor: variable length, to 32 characters—24 for v. 1.08 and below.

The descriptor field is the text descriptor for the device in alarm, as configured in the Pertronic Fire Alarm control panel. It identifies the location of the device in alarm.

6.2.4 Example:

<TAB>U900/4/Canteen West Wing<CR>	- not using the optional Panel field.
<TAB>U900/4/Bdg1 Canteen West Wing<CR>	- using the optional Panel field “Bdg1”.
<TAB>U900/4/Test Only<CR>	- test message example

6.3 Pertronic Network Pager Interface - AUSTCO

The AUSTCO protocol is used by the Pertronic Network Supervisory card when configured as an AUSTCO Pager (as opposed to AUSTCO_APR), to drive a paging base.

The communications parameters are as follows:

Type:	RS232
Baud Rate:	Configurable - 300, 600, 1200, 2400, 4800, 9600, or 19200 (default = 9600)
Data Bits:	8
Stop Bits:	1
Parity:	None
Handshaking:	None

The default structure of the message (2 lines) is:

<4 spaces>Time<space><hour(2 chrs)>:<minute(2 chrs)><4 spaces>Date<space><day(2 chrs)>-<month(2 chrs)>-<year(2 chrs)><CR>

3<3 spaces>FIRE:<building(3 chrs)><space><descriptor(9 chrs)><space><pager number(3 chrs)><CR>

The trailer is always a carriage return character, which will not print.

6.3.1 Building: 1 to 3 optional characters

Each Fire Control Panel has a dedicated network interface card, which is configured with its own descriptor. If the descriptor is configured (within the Pertronic Network Utility program) to end with **\xxx** where xxx can be up to three alphanumeric characters, then the Building field will be inserted using these characters.

6.3.2 Pager Number: 1 to 3 optional characters

The Pager Number is a 3 character configurable ID field in the Pertronic Network Configuration Utility for the network pager card.

6.3.3 Descriptor: variable length, up to 9 characters.

The descriptor field is the text descriptor for the device in alarm, as configured in the Pertronic Fire Alarm control panel. It identifies the location of the device in alarm. Note that the descriptor in the panel may be up to 32 characters, but only the first 9 characters are sent to the pager base.

6.3.4 Example:

Time 13:42 Date 23-08-08

3 FIRE:ABC Canteen 123

- where ABC is the optional Building field and 123 is the Pager Number

Time 13:00 Date 20-12-08

3 FIRE:Test Only 900

- test message example

6.4 KEW Pager Format

Three types of pager alarm are transmitted by the network supervisory card configured as a KEW Pager type.

a) Smoke detector alarms (F120 only)
b) MCP, heat detector, or sprinkler alarms (F120), MCP (F100A)
c) Smoke and Heat detector pre-alarms.

The data is sent at 300 to 19200 bps (selected on the Network Utilities package), 8-bit data, no parity, 1 stop and start bit.

CR as used below represents the ASCII carriage return character (0x0d in hex).

The messages sent to the equipment controlling the pager are:

a) Smoke Detector Alarms (F120):

"ALARM (SMOKE) " + [device descriptor to 32 characters]{ +" AT" +[panel name to 32 characters] + }CR

b) All Other Alarms (Generic):

"ALARM " + [device descriptor to 32 characters]{ +" AT" +[panel name to 32 characters] + }CR

c) Smoke Detector Pre-alarms:

"PRE ALARM (SMOKE) " + [device descriptor to 32 characters]{ +" AT" +[panel name to 32 characters] + }CR

d) All Pre-Alarms cleared (20 seconds delay)

"PRE-ALARMS RESET "

e) All Fire Alarms cleared (20 seconds delay)

"FIRE ALARMS RESET "

Optional exclusion of panel name:

The portion of the message enclosed by the {} brackets is not included in the message sent to the pager if the zone associated with the alarm is in a range selected on the Network Utilities package when the Pager configuration is set up.

Test Message:

Format:

ALARM<space>Test Only<CR>

Example:

ALARM Test Only<CR>

No panel field is included.

The device descriptor is hard coded to "Test Only".

7. PRINTER INTERFACE

The printer interface is available in version 3.0 firmware and above.

The Network Configuration Utility is used to configure the relevant operating parameters for the printer system - refer to the Network Configuration Program for details. The event queues to be printed are selectable from, alarm, network input, pre-alarms and isolate queues. Network defect, panel defect total, and system events are all printed by default. All events which are reported in NCU / NDU queues can be sent to the printer.

The printer interface connects to a serial printer through the network cards RS232 port using a null modem cable.

7.1 Message Format

First line

<day(2chrs)>/<month(2chrs)>/<year(2chrs)><space><hour(2chrs)>:<minute(2chrs)>:<second(2chrs><space><space><event description><CR><LF>

Second line

<4 spaces><descriptor if present><space>:<space><panel descriptor><CR><LF>

ASCII carriage return (CR – 0x0d) line feed (LF – 0x0a) characters are sent after each line.

7.2 Message Example

20/08/08 16:35:25 PANEL FAULT CHANGE Fault Count 002
<Panel descriptor>

20/08/08 16:35:25 ALARM (SMOKE) Zone 001 L01D001
<L01D001 Device descriptor> : <Panel descriptor>

Descriptors have maximum 32 characters.

7.3 Test Message

A Test message can be configured to be printed up to 4 times per day. Available with firmware v3.02 or later and Network Utilities v2.09.03 or later.

26/05/06 15:53:27 ALARM Zone 000 L01D000
Test Only : (Panel 000)

No panel descriptor field is included.
The device descriptor field is hard coded to “Test Only”.

8. NETWORK CONFIGURATION PROGRAM

8.1 Overview

Up to 99 network cards may be connected together to form a Network System. Each network card is connected to either a Fire Control Panel (F100A or F120A) or a Display Unit (Network Control Unit [NCU], Network Display Unit [NDU], Mini-Mimic, Network LED Mimic, or Pager Interface). The network may consist of up to 32 F100A and/or F120A panels and any combination of remote displays.

The LED Outputs from any panel may be mapped by the Network System to activate Network Inputs of other panels or LEDs of Network LED Mimics. For Fire Control Panels, these network inputs can then be configured to activate any desired outputs on that panel, such as output modules, local LEDs, logic blocks etc. For network LED mimics, the network input number relates to the LED number on the mimic itself.

The Network Configuration Program, running on a PC, defines the address and use for each of the network cards in the system, and the mapping of the LED Outputs of each panel to Network Inputs of other panels or LEDs of Network Mimics.

Network LED Mimics support 128 Network Inputs, which relate to LEDs 1 to 128 on that mimic respectively.

When used with the LED Address Controller, the base starting LED number can be selected.

When the network configuration has been completed using the Network Configuration Program, the PC may be connected to the RS232 connector of any one of the network cards and programming of the cards can commence. All network cards may be programmed from the connection at one network location, providing all cabling between them is intact and continuous as a loop.

Network Mapping Limitations:

a) each Network Panel card has a limit of 64 network outputs.
b) the total number of events which can be mapped to one network card is 255.
c) F100A has 64 network inputs that can be used to trigger panel outputs
d) F120A and Network LED Mimics each have 128 network inputs that can be used to trigger panel outputs
e) an LED output from a Fire Control Panel cannot be mapped to more than one network input of another individual panel. It can however be mapped to a network input of each remote panel separately. This limitation can easily be overcome by mapping the network input at the receiving panel to several outputs, or a group, as desired.

8.2 Installation

Run the ‘Install’ program from the supplied media or location on the computer hard drive. This is usually supplied as an *.msi program, which can be double clicked to run. The new version of the Network Utility will be installed in the “C:\Windows\Pertronic Network” directory.

i) If upgrading the software and the previous version is no longer required it should first be un-installed. This is achieved by selecting **Remove** from the options available from the Install program. Note that there may not be a prompt to remove an older version if the new version is incompatible with the previous configuration files. Once the previous version is removed, the update can then be installed.

ii) If it is desired to retain the previous version as well as installing the new version, then the “C:\Windows\Pertronic Network” directory should first be renamed. This will prevent the installation program removing the existing program. IE Rename “C:\Windows\Pertronic Network” to “C:\Windows\Network V2.06” or something similar. Then run the installation program and select **Remove** as in step i), prior to installing the new version. Once the installation is complete, create a desktop shortcut to the exiting “Netset.exe” network utility program in the directory that you have created. This will enable the old version of the program to be retained and run from the Windows desktop.

The setup program guides the user through the installation procedure.

At the end of the installation procedure the Setup Finish box will be displayed.

Run the Pertronic Network program from the folder in which it was installed or the icon automatically installed on the desktop, to operate the **Network Configuration Program**.

The **Network Configuration Program** screen consists of four sections in two parts:

Upper Screen: displays the network configuration:

Upper Left: four Radio buttons provide the means to configure the separate network components

Upper Right: displays the configuration details for each network device as defined by the Radio buttons

Lower Screen: displays the network status:

Lower Left: displays a list of the devices on the network. . This is referred to as the 'Network Address Summary' window. Click the + (plus) sign to expand the list.

Lower Right: displays the network messages. This is referred to as the 'Data watch' window.

The bottom line displays:

- the COM Port used
- the status of the PC connection.
- the network address number of the connected card
- the card type and version
- the IP address of the PC.

8.3 Data Entry

Click on the Network Configuration Program icon, or run the program from the Start menu.

The default Pertronic Network Configuration screen will be displayed. An error is reported if a network card is not located automatically at the default Com Port.

Configuration information is entered in the upper window of the display. At first a blank Network Panel entry screen containing only the column headings is displayed. Field names which have an asterix after them mean the field is to be modified by a text string or number entered from the keyboard.

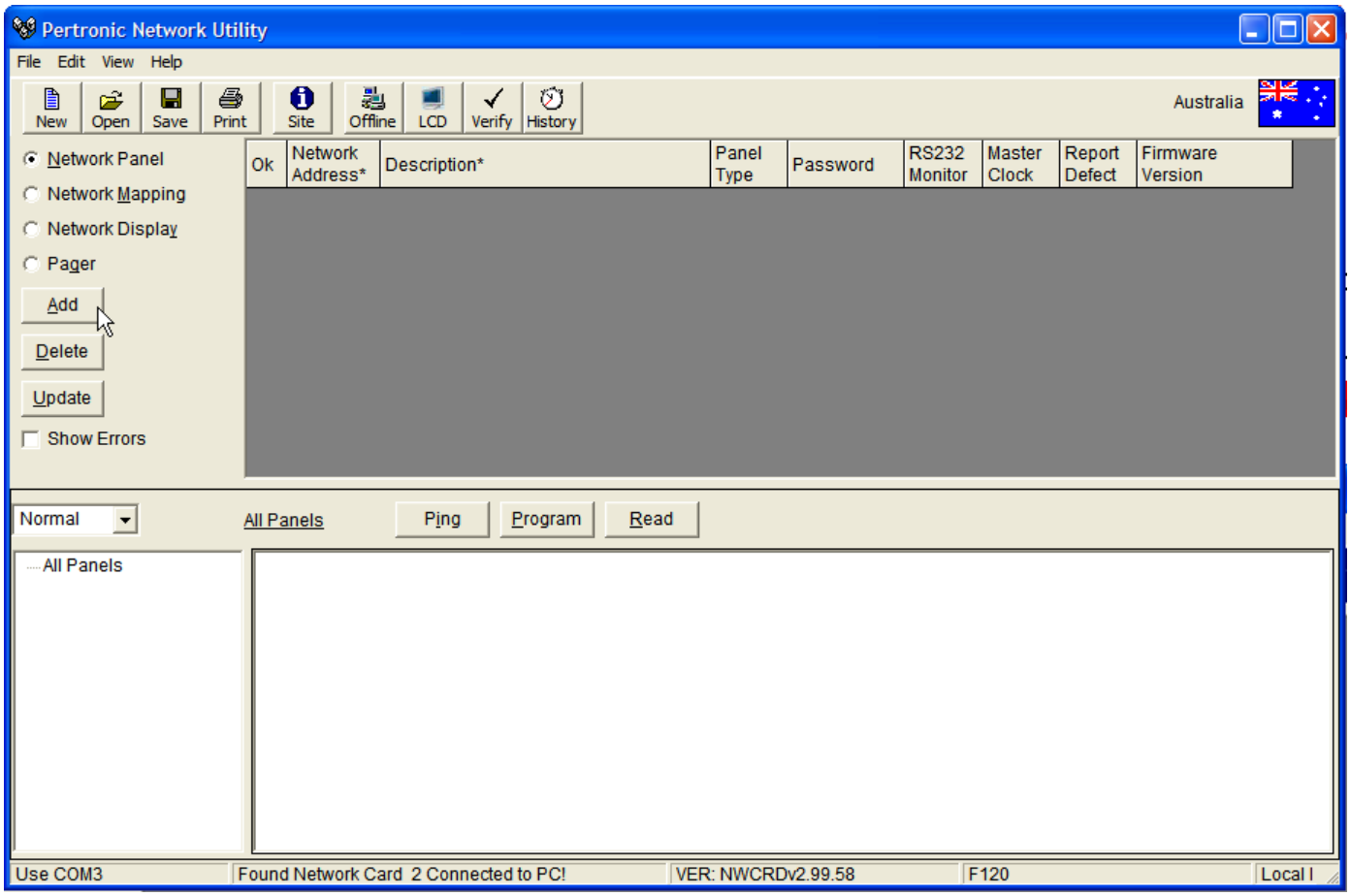


Figure 8.1: Network Configuration

Four Radio buttons, located at the top left of the Pertronic Network Configuration screen, provide the means to configure the network components:

Network Panels: configure network cards connected directly to panels

Network Mapping: configure LED to network input mapping between panels

Network Display: select and configure Network Display cards

Pager: select and configure Pager network cards

The flag in the top right hand corner of the screen shows the country used for the network configuration. This can be changed by selecting **Edit - Select Country** option from the **Edit** pull-down menu, then selecting the **New Zealand** or **Australian** radio button. See Figure 8.2 below.

Figure 8.2: Select Country

Incorrect selection may produce unexpected network results under certain conditions. A network defect will be reported on the network displays, and as a generic network error at the panels, if the panel card's configured country differs from the F100A or F120 panel country.

At any point during editing, the file can be saved to disk.

Press the **Save** icon on the toolbar or select **File-Save Config** or **File-Save Config As** in the **File** pull-down menu.

8.3.1 Connecting to Network

Click on the Connect to Network Card (**GO!)** Icon in the tool bar to bring up the Connection Options window (Figure 8.3). This may also be selected from the **Edit - Connection Options** field in the **Edit** pull-down menu. A drop down box can be used to select the com port to make the connection to the network. Press the Connect button to make the connection. If Auto Detect is selected clicking on the Connect button will scan all available Com Ports for the presence of a Network Card - this will return an error if none is found. When a network card is located, the com port, network card type, network number, and version are displayed at the bottom of the display, as in Figure 8.1: Network Configuration. Checking the **Connect On Start-up** tick box, will cause the Network Utility to attempt to auto-connect to the last used com port the next time the utility is started.

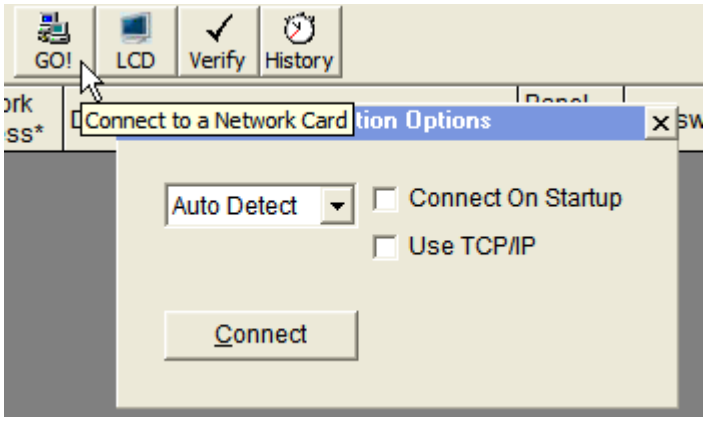


Figure 8.3: Connect Options

8.3.2 Network Site Information

The Network Utility program allows the user to enter site-specific information, which is saved in the configuration file but is not uploaded to the network. It is recommended that this be performed during the initial site programming as it preserves some important information.

To access the **Network Site Information** screen, select the site information icon in the task bar, or select **Edit-Network Site Info** from the **Edit** pull-down menu.

The fields available are shown in the figure below. Only text can be entered, limited to a single line for the **Site Name**, **Engineer** and **Date** fields, and several lines for the **Customer Information**, which could contain for example a full physical address of the site. Any data entered on this screen will also be included in a configuration printout, available through the **Printer** icon. It is however not stored on the network cards.

Any network diagnostic logs or history event logs generated will have the Site Name appended to the beginning of the file name to make them easier to identify.

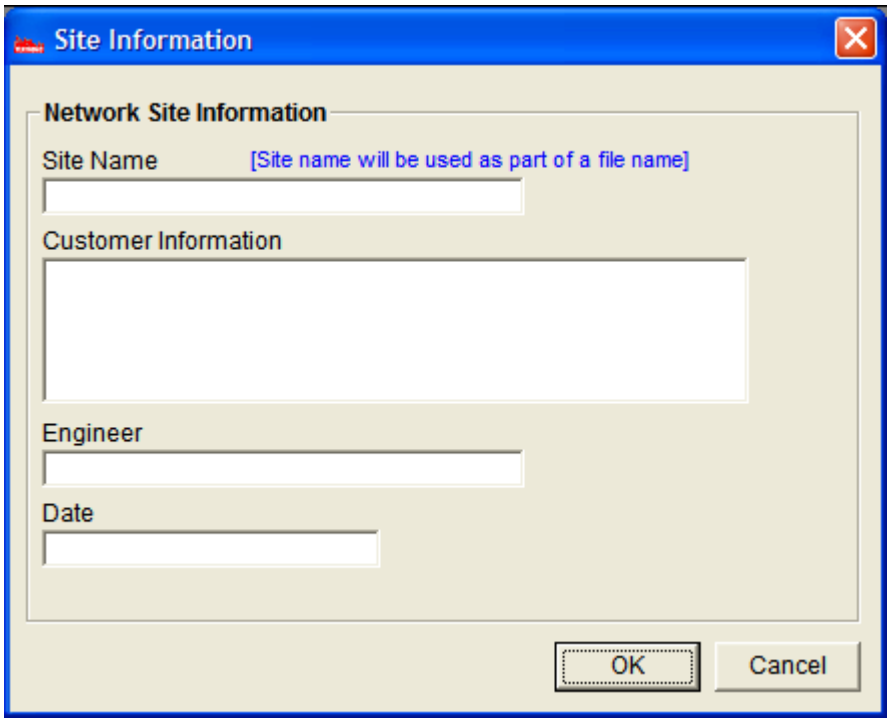


Figure 8.4: Network Site Information

The v2.08 Network Utility stores a portion of the **Site Name** on the network cards. Three characters of the Site Name are stored on each network card. Reading back a configuration from the network will automatically recreate the available portion of the **Site Name**. Subsequently saving a configuration read back from the network, will create a file with the Site Name appended to the beginning of the file name in order to aid future identification. Site Names may be manually altered at any time.

8.3.3 Network Panels

Click on the **Network Panel** button to select the network panel configuration data entry screen.

Panel Network Address (network card address 1 to 99), **Panel Type** (***F100A*** and ***F120A*** are currently supported), and the **Panel Description**, up to 32 characters (24 characters for firmware 1.08 or earlier), are entered here initially - Panel Number and Network Card Address are synonymous terms.

Note that a "\" character should not be used within the 'Panel Description', except for the AUSTCO or AUSTCO-APR Pager where the "\" functions as a Zone or Area Identifier – refer to AUSTCO and AUSTCO-APR:.

Include the panel access codes in the relevant column if one or more NCU displays are being installed. This then enables direct access to these control panels via the NCU. Access to the Engineering functions of the panel will still however require the access code to be manually entered by the user. An incorrect access code will inhibit remote connection from an NCU. Note that the access code is displayed as asterisks for security, and encrypted within the saved configuration file.

The **RS232 Monitor** field dictates if the RS232 connection between panel master board and the panel's network card is to be monitored. If NCUs are used on the system, a null-modem cable is required for this connection between the panel and its own network card, to allow the NCU to communicate directly with the panel in monitor mode. However, if no NCU is present, the cable serves no purpose and can be omitted and the RS232 Monitor field set to NO. Double click the field to toggle the setting.

The **Master Clock** field dictates which panel or network display (NCU or NDU) will control the Date and Time synchronisation for all displays and panels on the network, by sending out a signal once per minute - all displays which are out by 20 seconds or more will update their clocks. Only one panel or network display can be a Master Clock. Double click the field to toggle the setting to make for the relevant panel or display the Master Clock. Only the designated clock master or a network display can update the date and time for all connected displays. Usually a panel will be selected as the clock master, but an NCU or NDU can be configured as clock master in the case where all panels are headless F100 panels (a F100 panel with the main '*address 0*' display removed). In this case the panel does not have a real time clock so the network display must be used to synchronise network time.

Note: Polled mini mimics only, are able to be attached to headless F100 panels. A polled Full Function mimic is not able to be attached to a headless F100 panel as this will cause the network time updates to stop and disrupt normal F100 operation.

If an F120 panel is configured as clock master it will automatically update the network time for daylight saving (F100, NCU or NDU will not), thus it is recommended that an F120 is selected as a clock master where possible.

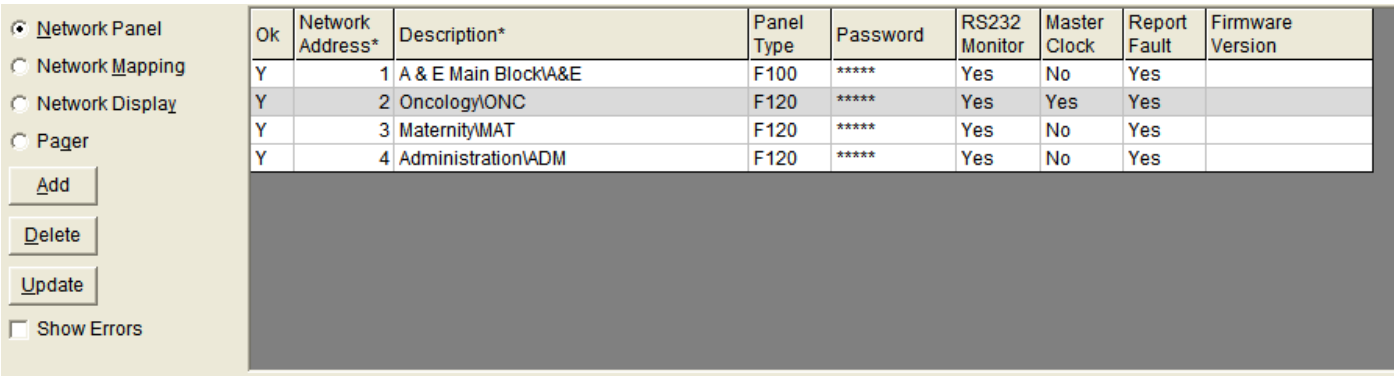
The **Report Fault** field dictates which panel or panels will indicate and report Network Errors. If each panel is able to individually generate a call to service engineers in the event of a defect, setting a single panel as a network fault reporter will limit the error to creating a single callout, as in the example below.

The **Firmware Version** field is not editable, and is simply used to show the firmware version of the relevant network card. This is completed automatically whenever the configuration is uploaded or downloaded to or from the network, or read from the network. It is important that all network cards contain the same firmware version to ensure compatibility, and a warning will be given if the program finds any differences. In particular, version 2.0 is totally incompatible with earlier versions. Version 2.0 is compatible with version 3.0 if the baud rate jumper AUX 1 is fitted on the version 3.0 card.

Press **Add** to add a new row to the Table. Click on the appropriate cell to enter or select the configuration information.

Press **Delete** to remove the currently selected row. Select the row by clicking on any one of the cells.

Press **Update** to refresh the rule checking for any newly added entries. If a rule has been broken '**Y**' will change to '**N**' in the rows **OK** field, and the **Show Errors** tick box will flash red. Checking the tick box will give a description of any errors in the **OK** field.



Ok	Network Address*	Description*	Panel Type	Password	RS232 Monitor	Master Clock	Report Fault	Firmware Version
Y	1	A & E Main Block\A&E	F100	*****	Yes	No	Yes	
Y	2	Oncology\ONC	F120	*****	Yes	Yes	Yes	
Y	3	Maternity\MAT	F120	*****	Yes	No	Yes	
Y	4	Administration\ADM	F120	*****	Yes	No	Yes	

Figure 8.5: Network Panel Configuration

8.3.4 Network Displays

Click on the **Network Display** button to select the Network Display data entry screen.

Press **Add** to add a new row to the Table. Enter the data into the corresponding columns of the Table by clicking on the appropriate cell.

Network Address: defines the network address (1 to 99) of the relevant network card. If the network card is required to interface with a pager as well as NDU or NCU displays, simply configure a Pager with the same address in the **Pager** edit window.

Description: allows a meaningful 32 character (24 characters for version 1.08 or earlier) descriptor for the relevant network card to be entered. This descriptor will be used for any network related messages from that card, such as a door being opened, or a network cable fault. In addition, NCU or NDU displays connected to that card will show the configured descriptor as their System Normal message.

Supervisory Type: Select from the available options, Universal NCU, NDU and LED Mimic. Graphics PC and Graphics & NCU are supported when using version 3.0 firmware only. These Supervisory types are not supported when using version 2.0 firmware.

Share Panel: indicates the default panel to which a Universal NCU will connect when the Panel Select button is pressed (not yet implemented). It is still possible to select a different panel however by scrolling through all available panels.

Number Connected: indicates the number of Displays (1 to 8) connected to the RS485 bus on the network card. Where more than one NDU is connected, the Display addresses must be set from 0 to the quantity minus one, with no gaps, to enable correct supervision of all Displays. If more than one Network LED Mimic is connected, the LED Mimic addresses must be set from 1 to the quantity (8 maximum).

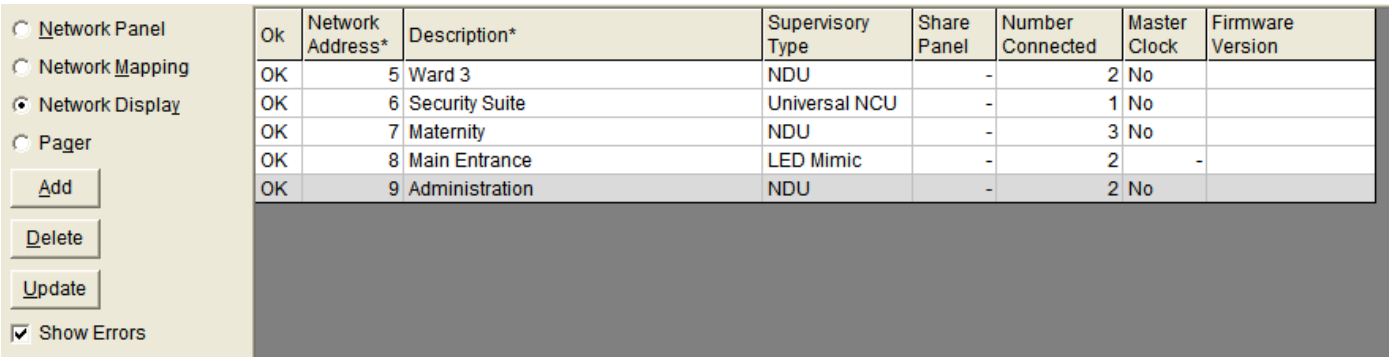
NCU displays must be the only display on a network card configured as a Universal NCU and must have RS485 address 1.

Master Clock: An NCU or NDU can be configured as the network master clock. This can be used in the case where all panels are headless F100s.

Firmware Version: is a non-editable field, and is simply used to show the firmware version of the relevant network card. This is completed automatically whenever the configuration is sent to the network, or read from the network. It is important that all network cards contain the same firmware version to ensure compatibility, and a warning will be given if the program finds any differences.

Press **Delete** to remove the current selected entry. Select the row by clicking on any one of the cells.

Press **Update** to refresh the rule checking for any newly added entries. If a rule has been broken '**Y**' will change to '**N**' in the rows **OK** field, and the **Show Errors** tick box will flash red. Checking the tick box will give a description of any errors in the **OK** field.



Ok	Network Address*	Description*	Supervisory Type	Share Panel	Number Connected	Master Clock	Firmware Version
OK	5	Ward 3	NDU	-	2	No	
OK	6	Security Suite	Universal NCU	-	1	No	
OK	7	Maternity	NDU	-	3	No	
OK	8	Main Entrance	LED Mimic	-	2	-	
OK	9	Administration	NDU	-	2	No	

Figure 8.6: Network Display Configuration

8.3.5 Pager

Click on the **Pager** button to select the pager data entry screen.

Press **Add** to add a new row to the Table. Enter the data into the corresponding columns of the Table by clicking on the appropriate cell.

Network Address: defines the network address (1 to 127) of the relevant network card. If the network card is required to interface with a pager as well as NDU or NCU displays, simply configure a Display with the same address in the **Network Display** edit window.

Description: allows a meaningful 32 character (24 characters for version 1.08 or earlier) descriptor of the relevant network card to be entered.

This descriptor will be used for any network related messages from that card, such as a network cable fault.

Pager Type: selects the pager to be connected.

- three options are available: **AUSTCO**, **AUSTCO-APR**, **KEW** and **PRINTER** (printer is only supported in firmware version 3.0 and above)**.**

Zone Start and Zone End: are relevant to the KEW pager only.

ID: identifies the pager interface to the Pager Base and is relevant to AUSTCO and AUSTCO_APR pagers only.

Baud Rate: may be selected from a list of standard rates – 300 bps to 19,200 bps.

The baud rate selected is used only for transmitting the Pager message. The receiver baud rate always defaults to 9600 bps.

Selected Queue: the events from any or all of the 'Alarm' (Fire), 'NW Input' or 'Pre-Alarm' queues may be selected to be sent to the pager by clicking on options within the Selected Queue field. The printer interface also has selectable 'Isolate' queue printing.

Test Option: enables test messages to be sent to the pager at pre-programmed times—whole hours only. Up to four messages can be sent each day. To select a test message time: click the Test Option field of the page interface, select the hour from the drop-down list, and check the "Test Msg at" box.

NOTE: The ALARM queue must be enabled to allow test message sending. The Test Option is only available with network firmware, v2.02 / v3.02 and above, in conjunction with Network Configuration Program v2.10 and above.

Firmware Version: is a non-editable field, and is simply used to show the firmware version of the relevant network card. This is completed automatically whenever the configuration is sent to the network, or read from the network. It is important that all network cards contain the same firmware version to ensure compatibility, and a warning is given if the program finds any differences.

8.3.5.1 AUSTCO and AUSTCO-APR:

Figure 8.7: Pagers shows the data entry table configured with both an AUSTCO and an AUSTCO-APR Pager. The options for these are identical, but the data format sent to the Pagers differs.

A ‘panel of origin’ descriptor, of up to 4 characters, is optional for AUSTCO-APR Pagers. This descriptor may be included in the ‘Panel Description’ field in the ‘Network Panels’ configuration data entry screen by appending a “\” followed by up to four description characters. This is shown in Figure 8.5: Network Panel Configuration for all the panel entries, e.g. Oncology\ONC, Maternity\MAT, etc., where 3 character descriptors are used.

The 32 character (24 characters for version 1.08 or earlier) ‘Pager Description’ field identifies the Pager interface to the other cards on the network.

8.3.5.2 KEW:

Figure 8.7: Pagersalso shows the data entry table with a KEW Pager configured.

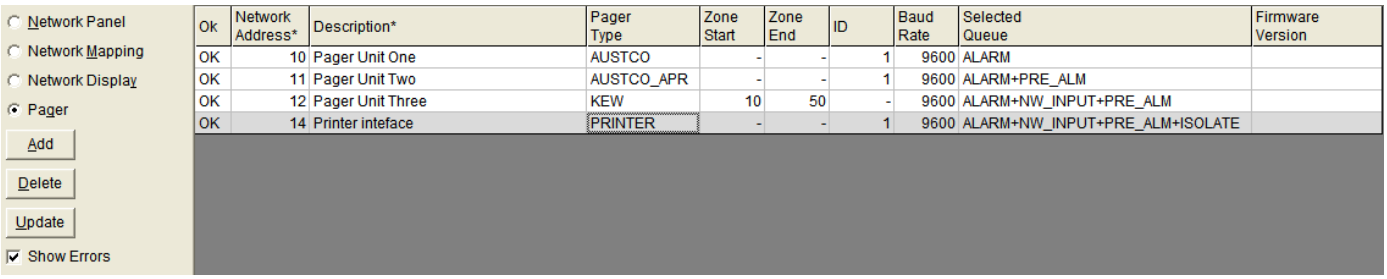
The **Zone Start** and **Zone End** fields (refer to Figure 8.7: Pagers) select an inclusive range of zones. If an event for the Pager has a zone number within this inclusive range, the panel descriptor component of the Pager message is discarded and is NOT sent to the Pager.

Notes:

a) Events within the zones selected, between **Zone Start** and **Zone End**, will be sent to the Pager with the **Panel Descriptor** text stripped out. Events from all other zones will have both “AT” and the ‘Panel Descriptor’ text from Figure 8.5: Network Panel Configuration, appended to the message sent to the Pager.

b) For the ‘Zone Select’ to operate over a networked site, each zone must have a unique number. Zone numbers must not be repeated in different sectors/buildings across a networked site.

Select the entry by clicking on one of the cells. Press **Delete** to remove the selected entry.



Ok	Network Address*	Description*	Pager Type	Zone Start	Zone End	ID	Baud Rate	Selected Queue	Firmware Version
OK	10	Pager Unit One	AUSTCO	-	-	1	9600	ALARM	
OK	11	Pager Unit Two	AUSTCO_APR	-	-	1	9600	ALARM+PRE_ALM	
OK	12	Pager Unit Three	KEW	10	50	-	9600	ALARM+NW_INPUT+PRE_ALM	
OK	14	Printer inteface	PRINTER	-	-	1	9600	ALARM+NW_INPUT+PRE_ALM+ISOLATE	

Figure 8.7: Pagers

8.3.6 Printer

Printer is only available with version 3.0 and above firmware.

Click on the **Pager** button to select the printer data entry screen.

Press **Add** to add a new row to the Table. Enter the data into the corresponding columns of the Table by clicking on the appropriate cell.

Network Address: defines the network address (1 to 99) of the relevant network card. If the network card is required to interface with a printer as well as NDU or NCU displays, simply configure a Display with the same address in the **Network Display** edit window.

Description: allows a meaningful 32 character descriptor of the relevant network card to be entered. This descriptor will be used for any network related messages from that card, such as a network cable fault.

Pager Type: select PRINTER as the pager type.

Zone Start and Zone End: not relevant to printer.

ID: not relevant to printer.

Baud Rate: may be selected from a list of standard rates – 300 bps to 19,200 bps.

The baud rate selected is used for transmitting the printer messages.

Selected Queue: the events from any or all of the ‘Alarm’ (Fire), ‘NW Input’, ‘Pre Alarm’ or ‘Isolate’ queues may be selected to be sent to the printer by clicking on options within the Selected Queue field.

Test Option: enables test messages to be sent to the pager at pre-programmed times—whole hours only. Up to four messages can be sent each day. To select a test message time: click the Test Option field of the page interface, select the hour from the drop-down list, and check the “Test Msg at” box.

NOTE: The ALARM queue must be enabled to allow test message sending. The Test Option is only available with network firmware, v2.02 / v3.02 and above, in conjunction with Network Configuration Program v2.10 and above.

Firmware Version: is a non-editable field, and is simply used to show the firmware version of the relevant network card. This is completed automatically whenever the configuration is sent to the network, or read from the network. It is important that all network cards contain the same firmware version to ensure compatibility, and a warning is given if the program finds any differences.

8.4 Mapping

Fire Panel LED Output to Network Input mapping can be configured in the Network Utility. The Table defines which events are mapped to LED Outputs at one panel, how these are mapped to Network Inputs at another, and what those network inputs will activate at the remote panel. Note that the Panel to LED Output, and Network Input to Output functions must be configured separately using the relevant panel Utility programs. These optional fields are included in the Table to simplify the planning and documenting of the mapping in the Network Configuration program. They are not uploaded to the network cards but are saved on the PC together with the configuration data.

Click on the Network Mapping button in the Network Configuration program to select the Network Mapping data entry screen.

Press **Add** to begin or to add new rows to the Table.

Enter the data into the corresponding columns of the Table by clicking on the appropriate cell. The example in Figure 8.8: Network Mapping shows outputs from 2 panels mapped to activate LED’s on the Network LED mimic.

The mapping entries can be sorted by source or destination panel in increasing order by clicking on the FROM NETWORK ADDRESS or TO NETWORK ADDRESS column descriptions. Clicking on the LED OUTPUT or NETWORK INPUT column descriptor will sort the output LED number or network input number for each panel in increasing order.

Input Comment: is an optional field provided to indicate the input function in the panel which will activate the LED, such as a Zone, Group, System Event, Detector address etc. The default contents “Optional Text” can simply be deleted and replaced with user text.

Description: is the descriptor of the network card which is being mapped to or from. This is automatically inserted by the program.

Panel Type: is automatically inserted by the program, and will reflect the type of panel configured at the network address in the **FROM NETWORK ADDRESS** field.

FROM NETWORK ADDRESS: is the address of the network card connected directly to the control panel, which is activating an LED, to be mapped.

LED OUTPUT #: is the LED being activated by the **FROM NETWORK ADDRESS** Panel.

NETWORK INPUT #: is the Network Input number to be activated at the receiving panel or mimic as a result of the LED activation previously configured. Note that for a Network Mimic, this is the actual LED number to be activated.

TO NETWORK ADDRESS: is the address of the network card connected directly to the control panel or LED Mimic, which is to act on the LED output activation previously configured.

Panel Type: is automatically inserted by the program, and will reflect the type of panel configured at the network address in the **TO NETWORK ADDRESS** field.

Panel Action: is an optional field, provided to indicate the output function of the receiving panel when the Network Input is activated, such as activating a particular loop module for example. The default contents “Optional Text” can be deleted and replaced with user text.

Note – Mapping Limitations

a) each Network Panel card has a limit of 64 network outputs.
b) the total number of events which can be mapped to any one network card is 255.
c) F100A has 64 network inputs that can be used to trigger panel outputs
d) F120A has 128 network inputs that can be used to trigger panel outputs

e) a network LED mimic has 128 network inputs that can be used to trigger mimic LEDs

f) an LED output from a Fire Control Panel cannot be mapped to more than one network input of another individual panel. It can however be mapped to a network input of each remote panel separately. This limitation can easily be overcome by mapping the network input at the receiving panel to several outputs, or a Group, as desired.

Select the entry by clicking on any one of the cells in the row. Press **Delete** to remove the selected entry.

Press **Update** to have the program automatically recheck the mapping rules. If a rule has been broken '**Y**' will change to '**N**' in the rows **OK** field. If any rule has been broken the **Show Errors** tick box will flash red, checking the tick box will give a description of the error in the **OK** field. See Figure 8.9: Mapping Rule Check.

Pressing update will also refresh the automatically generated descriptor and panel type fields.

Network Panel
Network Mapping
Network Display
Pager
Add
Delete
Update
Show Errors

Ok	Input Comment*	Description	Panel Type	FROM NETWORK ADDRESS*	LED OUTPUT*	To	NETWORK INPUT*	TO NETWORK ADDRESS*	Panel Type	Description	Output Comment*
Y	Zone 1	Panel 1	F100	1	1	==>	1	8	LED Mimic	Net Mimic Display	Index LED 1
Y	Zone 2	Panel 1	F100	1	2	==>	2	8	LED Mimic	Net Mimic Display	Index LED 2
Y	Zone 3	Panel 1	F100	1	3	==>	3	8	LED Mimic	Net Mimic Display	Index LED 3
Y	Zone 4	Panel 1	F100	1	4	==>	4	8	LED Mimic	Net Mimic Display	Index LED 4
Y	Optional Text	Panel 2	F100	2	1	==>	5	8	LED Mimic	Net Mimic Display	Optional Text
Y	Optional Text	Panel 2	F100	2	2	==>	6	8	LED Mimic	Net Mimic Display	Optional Text
Y	Optional Text	Panel 2	F100	2	3	==>	7	8	LED Mimic	Net Mimic Display	Optional Text
Y	Optional Text	Panel 2	F100	2	4	==>	8	8	LED Mimic	Net Mimic Display	Optional Text

Figure 8.8: Network Mapping

Network Panel
Network Mapping
Network Display
Pager
Add
Delete
Update
Show Errors

Ok	Input Comment*	Description	Panel Type	FROM NETWORK ADDRESS*	LED OUTPUT*	To	NETWORK INPUT*	TO NETWORK ADDRESS*	Panel Type	Desc
NW_Input Limit:Must be within the range 1-128 for a F120 or Led Mimic	Zone 1	Panel 1	F100	1	1	==>	200	8	LED Mimic	Net M
OK	Zone 2	Panel 1	F100	1	2	==>	2	8	LED Mimic	Net M
OK	Zone 3	Panel 1	F100	1	3	==>	3	8	LED Mimic	Net M
OK	Zone 4	Panel 1	F100	1	4	==>	4	8	LED Mimic	Net M
OK	Optional Text	Panel 2	F100	2	1	==>	5	8	LED Mimic	Net M
OK	Optional Text	Panel 2	F100	2	2	==>	6	8	LED Mimic	Net M
OK	Optional Text	Panel 2	F100	2	3	==>	7	8	LED Mimic	Net M
OK	Optional Text	Panel 2	F100	2	4	==>	8	8	LED Mimic	Net M

Figure 8.9: Mapping Rule Check

Select the **View-Mappings by Network Card** option in the **View** menu to bring up the network mapping summary window. See Figure 8.10: Mapping by Network Card. This allows each source and destination card's mapping entries to be summarised and checked.

A source card can be selected from the drop down box on the top left of the window. Any LEDs which have been mapped from this card will be displayed graphically in the window below this. The total number of panel output LEDs mapped from this card, and the total number of outputs mapped from all cards are displayed in a summary table below the window. In the bottom left hand corner is a summary of the source card's mapping entries. With the source card's panel output LED number and the corresponding destination address and network input for each entry.

Moving the mouse over a LED entry in the graphics display will highlight it and display the corresponding destination card address and network input number for the output LED. This is displayed in the bottom right hand corner of the window.

A destination card can be selected from the drop down box located on the top right of the window. All network inputs which have been configured for this card are shown graphically in the window below this. The total numbers of network inputs used by the card as well as the total number of network mappings to this card are displayed in a summary table below the window.

Moving the mouse over a Network Input entry in the graphics display on the right will highlight it and display the corresponding source panel address and output LED number for this mapping entry. This is displayed the bottom right hand corner of the window.

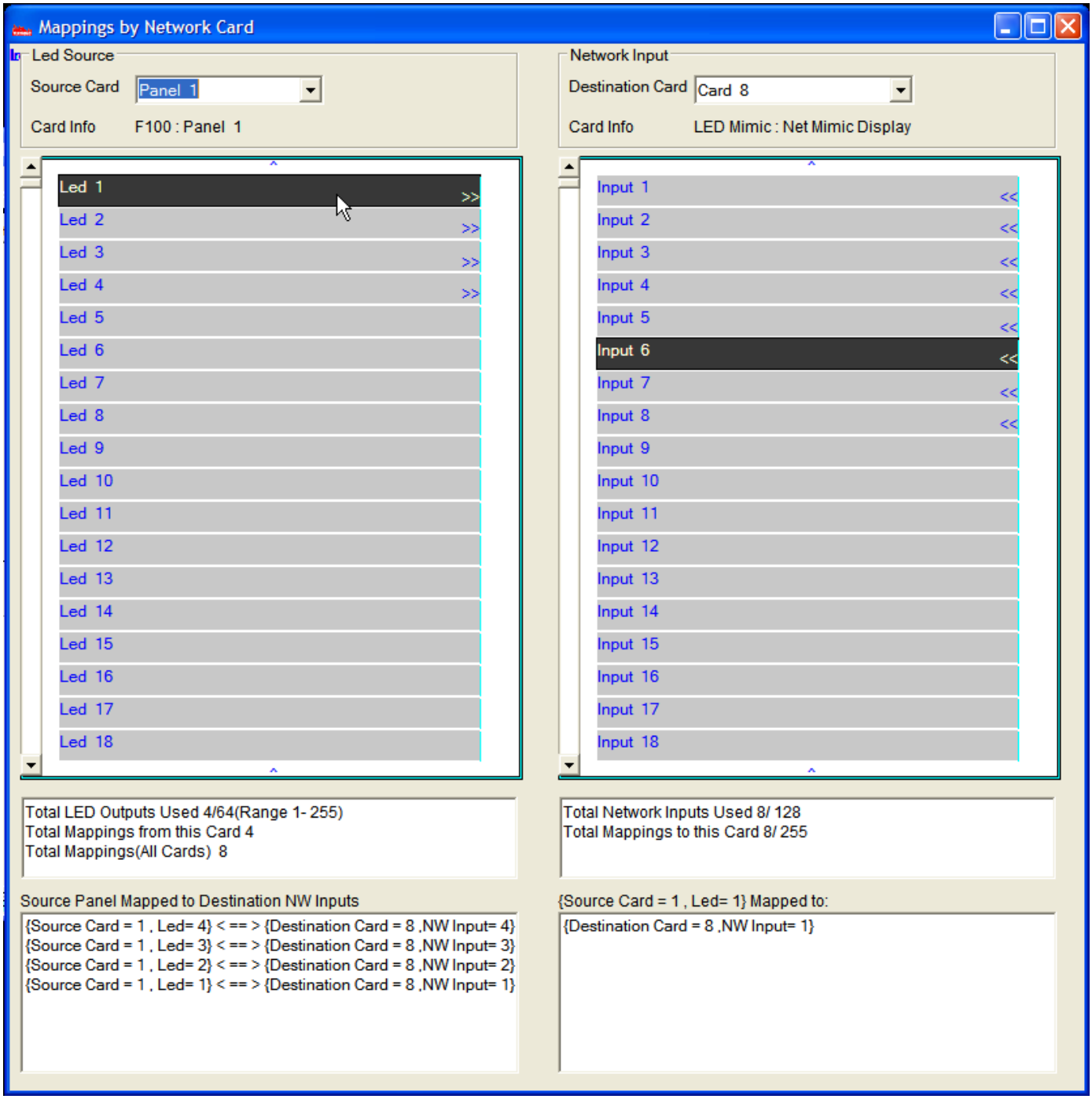


Figure 8.10: Mapping by Network Card

8.5 Configuration File Transfer

8.5.1 Select Connection Setup

When the Configuration program is first run selecting the Connect to Network card Icon **(GO!)** from the tool bar will bring up the Connect Options window. There is a drop down box which can be used to manually select the com port to use for the connection. The default option is Auto Detect, if this is selected, pressing **Connect** will automatically find a com port which has a network card connected.

It is advisable to select the Auto Detect option only when a network card is already connected and powered up.

The D9 PC to Network cable may be connected to any one of the system network cards at the RS232 connector. The configuration for the network cards can be uploaded, and the existing network configuration can be downloaded and/or verified against a file copy from that single point.

The PC cable used is a Null-Modem cable, identical to that used to configure other Pertronic Analogue Addressable control panels.

8.5.2 Uploading the Configuration

When data entry for the Network Panel, Network Display and Network Mapping and Pager sections is complete, the configuration may be uploaded to the network cards of each of the system panels and interfaces.

> **NOTE**: For the configuration upload process to take place, the network cable loop **must** be complete and the network loop communications operational.

The lower left window of the Network Configuration screen contains a list of the devices on the network.

- select the **+** sign next to **All Panels** to view the individual panels and network cards.
- select **All Panels** (default) to program all the network cards together, or select individual cards to program them individually. Normally **All Panels** will be selected.
- the **Normal/Debug** drop-down list (top left) is set to Normal by default. If required for diagnostic purposes, it can be set to **Debug**.
- press **Program** to start the upload process. The status of the configuration upload is reported in the lower right hand window of the Network Configuration screen. During the data transfer, all panel network cards are automatically isolated, restricting data traffic between panels and the network. Panels display the isolation during this process.

The programming sequence is:

a) the network cards firmware versions are read, and an error raised if they are not all the same version. Programming can optionally continue however where possible. If there are differences, it is strongly recommended that relevant cards be upgraded. Note that firmware version 2.0 is incompatible with earlier versions. Version 3.0 is compatible with version 2.0 if a baud rate jumper is fitted to board with version 3.0 firmware (PCB v2.20).
b) the network cards are reset, clearing any previous configuration.
c) the network cards are then configured with the type of panel connected and other parameters set.
d) the network cards are sent the mapping configuration - panel LED Number to other panel Network Input.

If an upload to a particular network address fails, this is displayed in the Program window with details of the error written under the failed command and **Programming completed with Errors** in red underneath.

In **Error! Reference source not found.**, it is clear that the network cable is incomplete, i.e.: not in a closed loop.

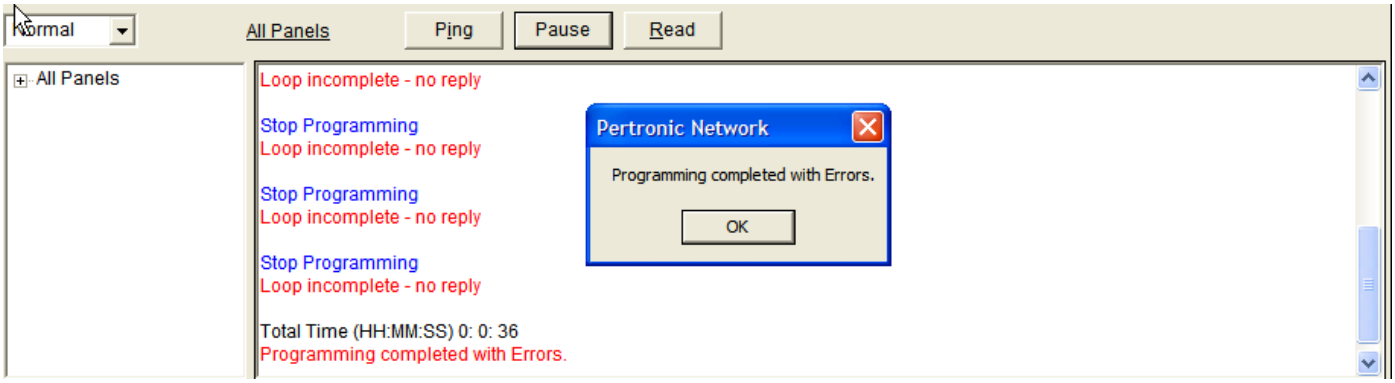


Figure 8.11: Upload Error

If errors occur, check that the network address is valid and that the network card is operating correctly. If the problem persists, contact Technical Support.

If there is a discrepancy in the network card firmware, the option to continue may be provided - select Yes to continue. It is strongly advised that all network cards are at the same version to ensure compatibility.

Note that no such option is given if any network card is at version 2.0 or later AND any other card is at version 1.08 or lower.

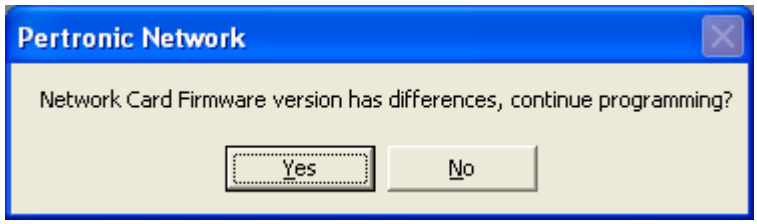


Figure 8.12: Firmware Version Difference

If all commands are carried out successfully, a screen such as Figure 8.13: Upload Success is displayed. The full upload history can be viewed by scrolling up or down the window using the slider bar on the right.

Every command should have **OK** underneath to show that the relevant network card received it correctly.

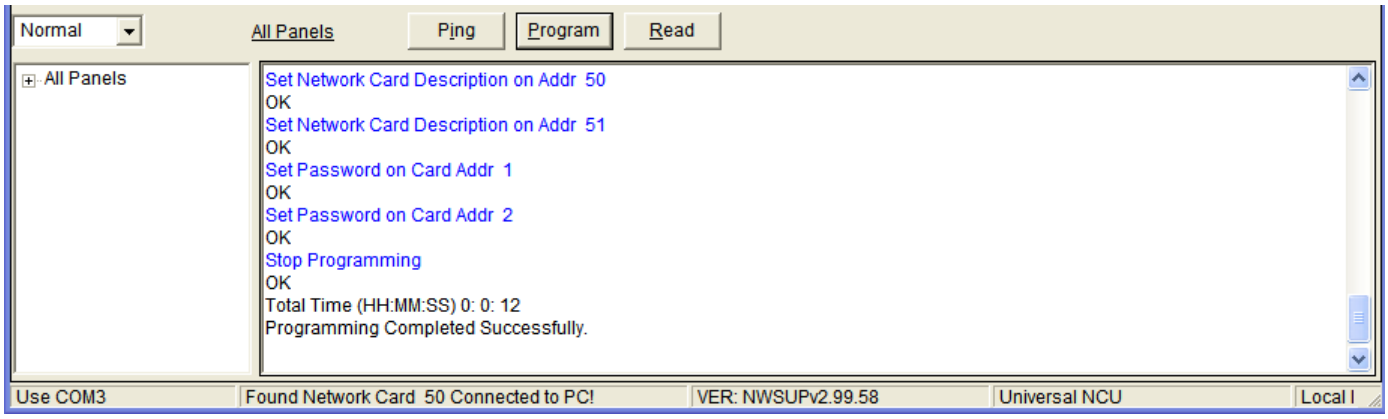


Figure 8.13: Upload Success

8.5.3 Downloading the Configuration

The configuration held by all the network cards can be read (downloaded) by the Utility, and then saved onto disk if required, by selecting the **READ** button. The data is then downloaded from the network, and the relevant pages of the configuration display the contents of the network card configurations. This data can be saved as a configuration file on the PC using the **Save Config** or **Save Config As** menu options. Note that the optional fields in the **Network Mapping** screen and the **Network Site Information** will be blank as this information is stored on the local disk file and not in the network cards. The firmware versions of all cards are also downloaded and any discrepancy is reported. It is strongly advised that all network cards are at the same revision.

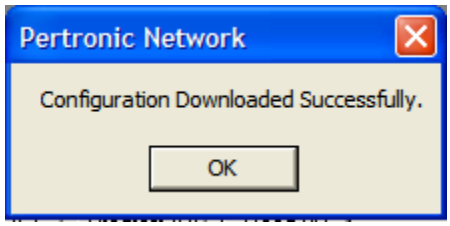


Figure 8.14: Download success

8.5.4 Verifying the Configuration

Use the **View-Verify Configuration** menu option to verify (compare) the current network configuration with one saved on the PC. To do this, it is first necessary to open an existing configuration file, then select the **View** menu, and select the **Verify Configuration** command. The data will be downloaded from the network cards and displayed on screen as done with the **READ** option. The Utility then compares the newly read-in data with that which was loaded into the program, and a verification report is produced if there are any differences, clearly highlighting the discrepancies **in bold type**, and saving a copy of the report to disk for future reference. Note the location of the file at the bottom of the report. See Figure 8.16: Verification Report.

If the verification process finds no differences, a message will report this fact. Be aware that if the file is now saved, it represents the true configuration of the network and not the file which had first been loaded into the program. The user is prompted for a filename to save it under.

Note that the optional fields in the **Network Mapping** screen and the **Network Site Information** are blank as this information is stored on the local disk file and not in the network cards. The firmware versions of all cards will also be downloaded and any discrepancy reported. It is strongly advised that all network cards are at the same revision.

Figure 8.15: Verification Okay

Print

Verification Report

Current Network Utility in use:v2.07.65

Date:30/04/2008 10:41:54 a.m.

Project	
Customer Information	
Engineer	
Date	

Network System Type :Australia

From Network Card Address: 50

Firmware Version Read Back (Network Card Connected to PC):NWSUPv2.99.58

Local File:2pan2sup_140405.pnt

*** Panel Configuration**

Configuration Item	Differences					
Panel2	Exists in Network, but not in configuration					
Address	Location	Type	Password	RS232 Monitor	Master Clock	Description
Panel1	Local	**F120**	*****	No	**Yes**	Panel 1
	Read Back	**F100**	?????	No	**No**	Panel 1

*** Supervisory configuration**

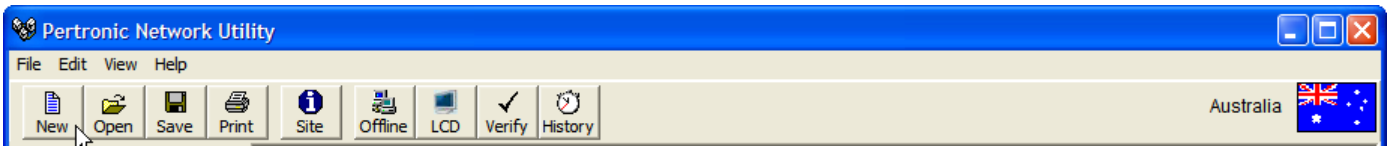
Configuration Item	Differences					
Address	Location	Type	Share Panel	Number Connected	Master Clock	Description
Supervisory51	Local	**Universal NCU**	-	**1**	No	NDU addr 51
	Read Back	**NDU**	-	**2**	No	NDU addr 51

A html file has been created at the following path:

C:\Pertronic_File\Verification\ Verification Report 2008-04-30 1041.html

Figure 8.16: Verification Report

8.6 Toolbar and Drop-Down Menus



Various options and functions are available via the menu system and/or icons at the top of the program screen:

8.6.1 File menu

Allows the user to:

New Config - create a new file

Open Config - obtain an existing file

Open Recent Config Files - choose from a list of recently used config files to open

Open Recent History Log Files - choose from a list of recently generated log files to open

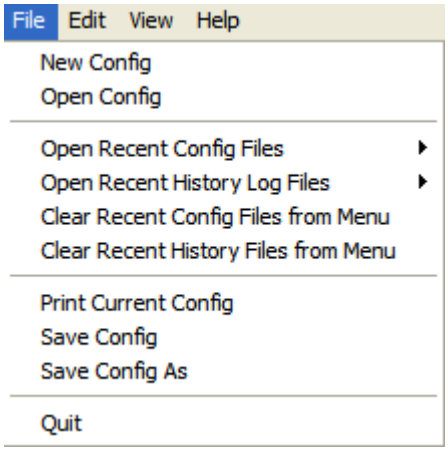
Clear Recent Config Files from Menu - resets list of recent config files

Clear Recent History Files from Menu - resets list of recent history files

Print Current Config - creates a network configuration report file which can be printed

Save Config and **Save Config As** - save files

Quit - exit the program



8.6.2 Edit menu

Allows the user to:

Network Site Info - bring up the network site information window.

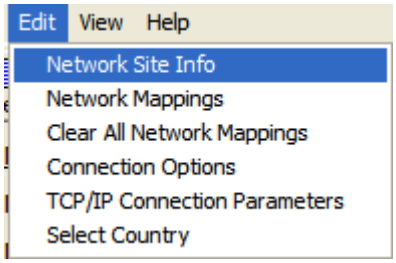
Network Mappings - edit the network mapping.

Clear All Network Mappings - clears all mapping in configuration.

Connection Options - brings up the network card connection options window.

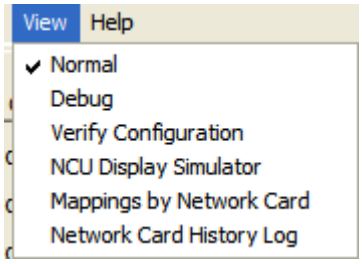
TCP/IP Connection Parameters - brings up the TCP/IP connection options window.

Select Country - brings up the select country window, for selection between New Zealand and Australian systems.



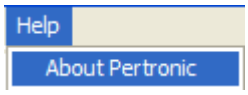
8.6.3 View menu

The **View** menu gives alternative access to the configuration mode (**Normal** and **Debug**) menu items, as well as access to the **Verify** configuration function, the **Network Card History Log**, (refer to Network Card Fault Log (section 9.1) and Network Card History Log (section 8.6.13) for details), and the **NCU Display Simulator** (see section 8.7). It also gives access to the **Mappings by Network Card** screen (section 8.4).



8.6.4 Help menu

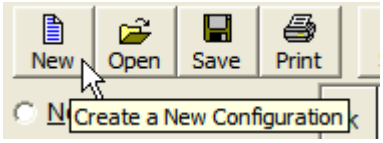
The **Help** menu gives access to the **About Pertronic** information, which gives the revision number of the program. This may be needed when requesting technical assistance for the Network system.



8.6.5 New Configuration icon

The **New** file icon creates a new network system configuration.

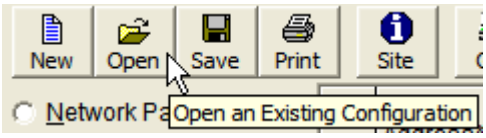
Previous Table entries are cleared



8.6.6 Open icon

The **Open** file icon opens and loads an existing configuration file.

Configuration files have the file extension: pnt.

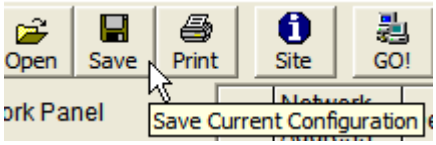


8.6.7 Save icon

The **Save** icon saves the current configuration file to disk.

A filename will be requested if this is a new configuration.

The file is saved with the file extension: pnt.



8.6.8 Print icon

The **Print** icon opens the default Browser program displaying the data from the configuration file. A copy of this printable html file is created in the folder containing the program, as listed at the bottom of the file. The file can then be printed from the browser.

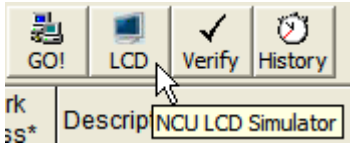
8.6.9 Site Information icon

The Site Information icon opens the network site information window.

8.6.10 Network Connection icon

Selecting the network connection icon opens the connection options window, which is used to select a com port and make a connection from the PC to the network card.

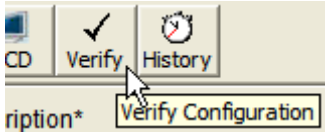
8.6.11 NCU LCD Simulator



Selecting the NCU LCD Simulator icon opens an LCD display simulator; this simulates the physical NCU's display on the PC, provided that PC's RS232 cable has been connected to a network supervisory NCU configured card.

See section 8.7 for details on operation.

8.6.12 Verify Configuration



Press this icon to verify an opened or just edited local configuration file with the network system configuration that has been loaded.

8.6.13 History Log

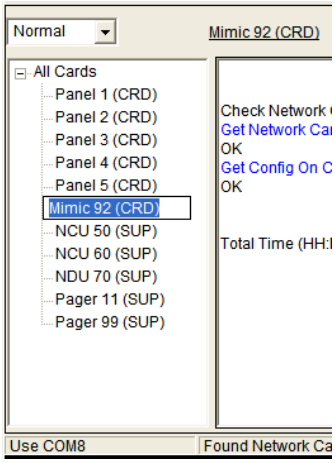


Figure 8.17: Network Address Summary

Press this icon to download the history log from the supervisory card that is connected directly to the PC. The history log contains a record of all network events. The number of events which are to be downloaded can be selected up to a maximum of 2000 events. Alarm, Isolate, Network Input and Pre Alarm queues can be filtered, so only events which are logged of this type are downloaded from the network card history log. Events for only a single panel, pager or supervisory display can also be filtered out, by first selecting the network address from the network address summary window.

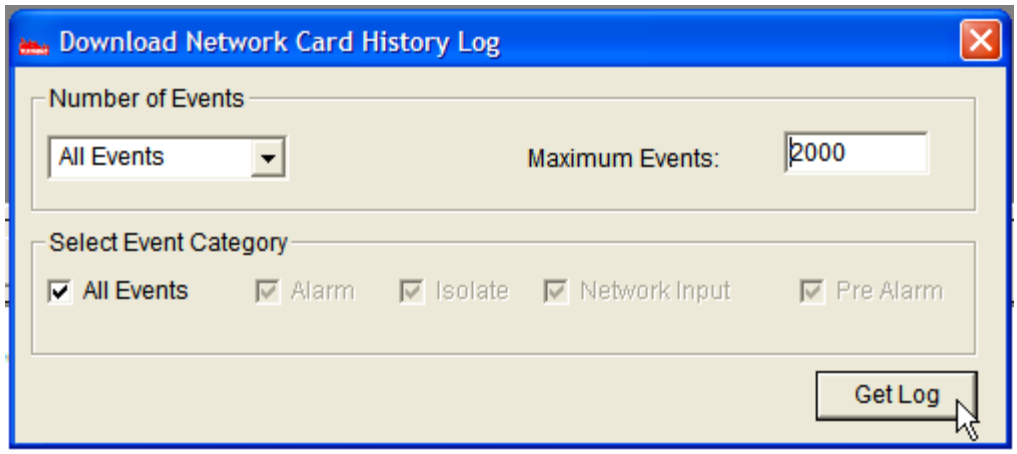


Figure 8.18: History Log

History log downloading is only supported with v3.0 firmware (v2.20PCB). If a history log download is attempted on V2.0 firmware (v1.09PCB) then the warning in Figure 8.19: Firmware Version Warning is displayed.

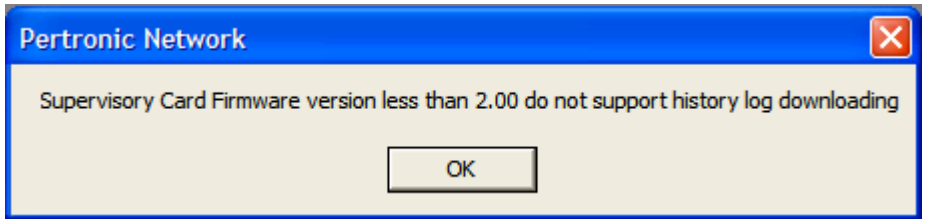


Figure 8.19: Firmware Version Warning

If connected to a panel card the panel card network fault log will be downloaded. This has a maximum of 128 events and does not allow filtering of downloaded event queues or the number of events downloaded to be changed. Fault logs contain network fault messages only. All other events such as Alarms and Isolates are not recorded.

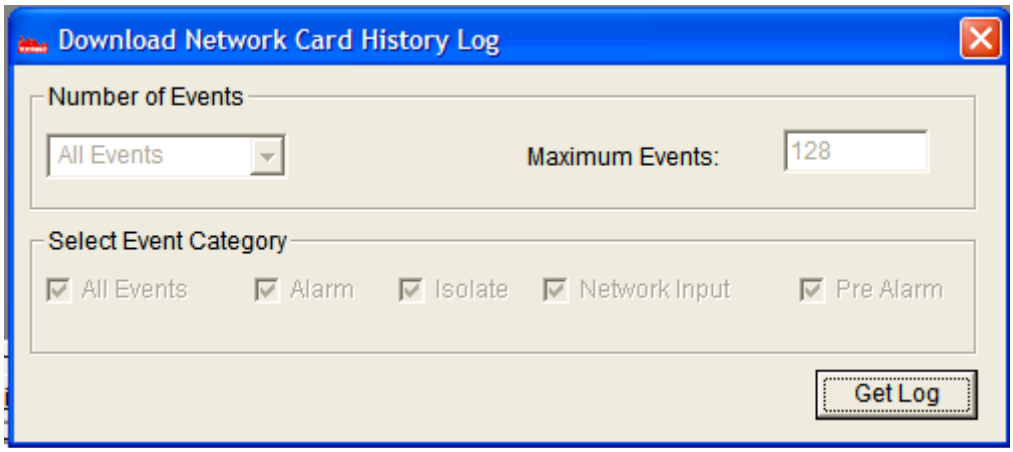


Figure 8.20: Panel Fault Log

Panel card fault logs are stored on the network card in volatile memory. Removing power or resetting of a panel card will clear any events stored in that cards fault log. Supervisory card history logs are stored in non-volatile memory and can only be erased by loading the network card with new firmware. Refer to section 2.2 for details of firmware versions.

8.7 NCU Simulator

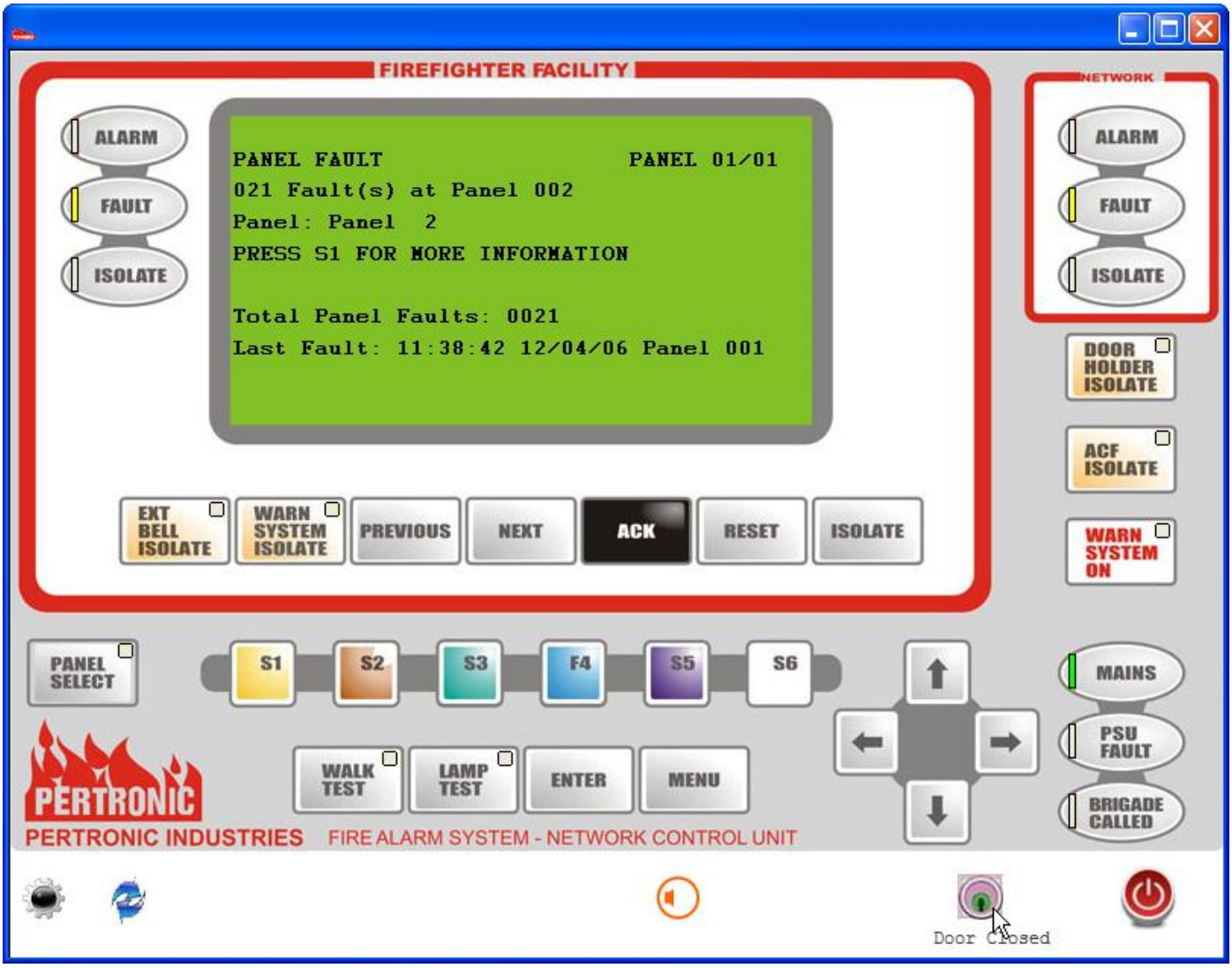
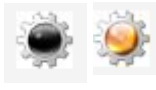


Figure 8.21: NCU Simulator

If the PC is connected to a network card which has been configured as an NCU, and the NCU Simulator icon is selected from the toolbar, or from the **View - NCU Display Simulator** menu option, the display above will be shown. The simulated display on the PC will mimic the real NCU connected to the network card. All control functions are available, including interrogation of any individual control panels, as described in Section 4.

In addition, there are 5 icons at the bottom of the window, described below:

8.7.1 Communication Status



This icon will occasionally blink to indicate communication between the simulator and the network card. The example shows the off and on indications respectively

8.7.2 Refresh Display

This icon can be clicked to force a refresh of the display in case communications between the simulator and the network card is lost for any reason.

8.7.3 Buzzer Status

This icon shows the status of the NCU buzzer, Off and On respectively in the examples. As with the NCU, the buzzer can be silenced by pressing any NCU control button, typically the ACK.

8.7.4 Close Window

Clicking on this icon will close the simulator screen window, reverting to the Network Utility

program screen. It will automatically close the simulator door if not already closed.

8.7.5 Door Status

This icon and message both control and display the status of the simulator's door. As with a real NCU, until the door is opened, no NCU control buttons are accessible. On clicking the icon, the door status will toggle between Open and Closed. This has the same function as the door switch on NCUs, NDUs and control panels, in that a Door Interlock fault will be reported when the last door is closed if an event such as 'Warning System Isolate' is active at the time.

8.8 Network Diagnostics

8.8.1 Network Ping

Press the **Ping** button to send network loop ping messages to each card in the network loop in both directions. This can be used to check the integrity of the network loop prior to programming. If the ping reply is received correctly OK will be printed, other wise "Loop incomplete - no reply" error message will be reported after a timeout period. See Figure 8.22: Network Ping. Individual cards can be pinged by selecting the card after clicking on the (+) plus in the network address summary window.

A network ping message is sent out of the source card's network loop In port or Out port to any of the cards in the network. The ping reply will come back from the destination card in the same route as the ping was sent. So a reply means the network loop up to the pinged card in the direction of the ping is complete.

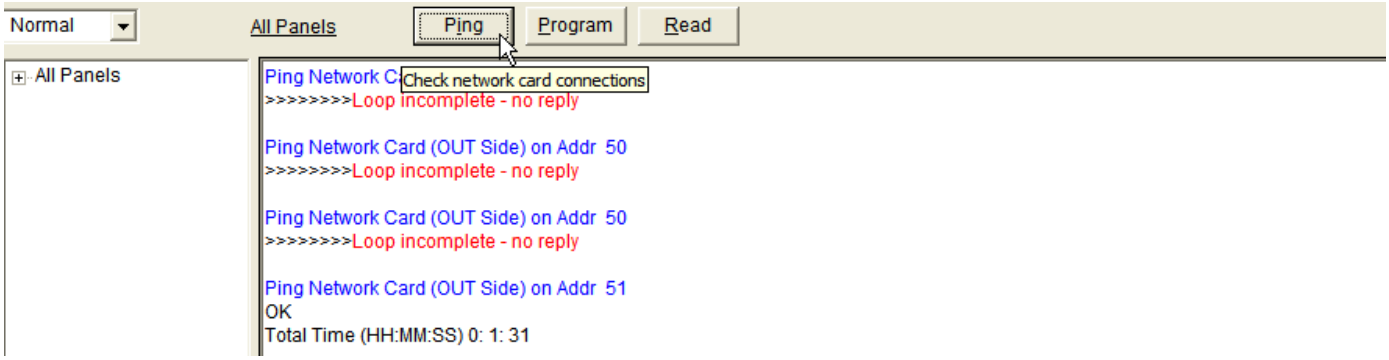


Figure 8.22: Network Ping

A Network Diagnostic Report is generated which shows a summary of the network ping attempts. See Figure 8.19: Network Diagnostic Report. A table is produced which has an entry for each card on the network which was pinged and its status. Each entry has the card network address number, the configured card type and its description.

The In Side field reports OK if the card could be pinged from the source card's network loop In port, or 'loop-incomplete' if no reply is received.

The Out Side field reports OK if the card could be pinged from the source card's network loop Out port, or 'loop-incomplete' if no reply is received.

The Out Side Missing Reply Count field displayed for each card is a counter of the number of messages sent from the card's output port which have not been acknowledged correctly. This may be used to determine if there is any intermittent problem between this card's output loop connection and the next card's Input loop connection.

Through Previous Card (X)'s In Side field returns the address of the card before the pinged card if the ping is sent in the In loop direction.

Through Previous Card (X)'s Out Side field returns the address of the card before the pinged card if the ping is sent in the Out loop direction.

Current Network Utility in use:v2.07.65

Date:30/04/2008 4:19:36 p.m.

Project	test net1
Customer Information	
Engineer	
Date	

Network System Type :Australia

From Network Card Address: 50

Firmware Version Read Back (Network Card Connected to PC):NWSUPv2.99.58

Network Address	Card Type	Card Description	In Side	In Side Retry Times	Out Side	Out Side Retry Times	Out Side Missing Reply Count	Through Previous Card (X)'s In Side	Through Previous Card (X)'s Out Side
2	F120	Panel 2	OK	0	Loop incomplete - no reply	3	12	50	0
50	Universal NCU	Universal NCU 50	Loop incomplete - no reply	3	Loop incomplete - no reply	3	0	0	0
1	F100	Panel 1	Loop incomplete - no reply	3	OK	0	0	0	50
51	NDU	NDU addr 51	Loop incomplete - no reply	3	OK	0	1290	0	1

A html file has been created at the following path:

C:\Pertronic_File\Diagnostic\test net1 Network Diagnostic Result 2008-04-30 1619.html

Figure 8.23: Network Diagnostic report

By using PING it is possible to confirm:-

- the existence of a network card
- the network wiring order
- any intermittent communication errors between network cards
- and the location of any single network break

In the above example the network wiring order is 50-1-51-2 and there is currently a loop break between network addresses 2 and 51. It is not necessary to configure network cards prior to using the PING command.

9. NETWORK MESSAGES

The Network System is designed to tolerate single loop communication failures, typically loop shorts, open circuits, or component failures, so that the system continues to operate normally. These failures and other network faults are reported to the connected panels.

Note that only the panel connected to the network card reporting the fault will display the type of network error; all other panels will report a generic Network Error message.

It is strongly recommended that at least one Network Display be installed (NDU or NCU) to provide a better view of network events.

Network errors are non-latching and clear automatically when the problem is resolved.

In all instances, a network error is displayed at an NDU or NCU together with the address or descriptor of the network card with the error.

Network errors and system messages are detailed in the table below:

F100 LCD Message	F120 LCD Message	NCU and NDU Message	Meaning
Network Dupl Addr	Network Duplicate Addr	Network Duplicate Address	Another network card on the system has the same address as the connected card. Change the address of one card to an unused address. A NW CARD LOST message may also be created, referring to the panel card whose address is set incorrectly, according to the configuration
Network Lost [IN]	Network Lost [IN]	Network Loop Broken Address XXX (IN) and YYY (OUT)	The communication link between the connected card and the network card connected to the IN terminal has been lost. Check the wiring and connectors for open or short circuits
Network Lost [OUT]	Network Lost [OUT]	Network Loop Broken Address XXX (IN) and YYY (OUT)	The communication link between the connected card and the network card connected to the OUT terminal has been lost. Check the wiring and connectors for open or short circuits
Network Earth Fault	Network Earth Fault	Network Earth Fault	The network cable connected to the OUT terminal of the relevant panel network card has a connection to Earth. Check the cable connection.
Network Error	Network Error	Network Earth Fault	The network cable connected to the OUT terminal of the relevant non-panel network card has a connection to Earth. Check the cable connection.
Network Error	Network Error	Network Card Lost	The non-panel network card reported by the NDU or NCU is no longer responding. It may have lost power, or is set to an incorrect address for the configuration
Net Card Missing	Network Missing	Network Card Lost	The panel network card is no longer responding. It may have lost power or data (RS485) connection to the panel
Network Error	Network Error	Remote Display Comms Error	A Display (NCU, NDU or Network LED Mimic) is no longer responding to its' own network card. Check power and data connections (RS485)
Network Error (Clear)	Network Error (Clear)	N/A	Indicates that the network error not associated with the connected card previously reported, has cleared.
Network Card Isolated	Network Isolated	Network Card Isolated	The isolate switch of the connected card is active. Not an error but shown in the Isolate queue
Network Error	Network Error	Panel Rs232 Connection Error	The RS232 null modem cable between the panel RS232 port and that panel's network card is not connected
Network Error	Network Error	Door Interlock Fault Function Isolated	All panel and supervisory display doors are closed whilst the global Silence Alarms button is active on the supervisory displays
Network Error	Network Error	Panel Connection Error	Network card is not receiving panel polls. Either problem with panel to network card RS485 connection or panel not enabled for networking. All other panels report a generic network error
Network Error	Network Error	Panel Type Error	Configured panel network card type (F100 / F120) does not match physical panel type. Only the local panel reports a generic network error
Network Error	Network Error	Panel Country Error	Configured panel network card country (Australia / New Zealand) does not match physical panel country. Only local panel reports a generic network error
Network Error	Network Error	NW Message Checksum Error	Detected checksum errors in network messages. Panel network cards will report a generic network error at their own panel.
Network Error	Network Error	Program/Config CRC Error	The configuration data or program code memory has been corrupted. A generic network error is sent to all panels.
Network Error	Network Error	Comms Error NDU XX	Displayed on all surviving NDUs connected to the same network card as the one with communications loss.

Table 9.1: Network Messages

9.1 Network Card Fault Log

In addition to the individual Fire Control Panels reporting and logging their own network card faults, the following errors are logged by each of the network cards directly connected to ***F100A*** or ***F120A*** Fire Control Panels, or **Network LED** mimics (not Network Display Cards, these contain a full network event history log) and can be accessed using the **View- Network Card History Log** menu option within the Utility or by clicking on the history icon in the tool bar after connecting to a panel card, to display the contents of the fault log in a user-friendly format.

A null-modem cable is used for the connection, the same cable used for the Network Configuration upload and download.

The Fault log is volatile, therefore a power failure, board reset or address change will clear the log.

The error log lists the event number, network card address reporting the error, and the time and date of the event, based on the network synchronised time.

An example of a Network Fault log obtained from a panel network card follows:

Print

Network Card Fault Log

Current Network Utility in use:v2.07.65

Date:1/05/2008 9:07:18 a.m.

Project	test net1
Customer Information	
Engineer	
Date	

Network System Type :Australia

From Network Card Address: 2

Firmware Version Read Back (Network Card Connected to PC):NWCRDv2.99.58

Num	Type	Address	Time	Day/Month
1	Lost Comms (OUT)	51	18:11:22	11/04
2	Lost Comms (IN)	2	18:11:20	11/04
3	Network Card Missing (cleared)	50	17:59:57	11/04
4	Network Card Missing	50	17:54:21	11/04
5	Lost Comms (OUT) (cleared)	51	11:02:46	12/04
6	Lost Comms (IN) (cleared)	2	11:02:45	12/04
7		0	00:00:00	00/00
8		0	00:00:00	00/00

Figure 9.1: Network Fault Log

The full list of raw Error codes logged is:

Network Error	Code
Duplicate Address	0
Duplicate Address Cleared	1
Lost Comms (IN)	2
Lost Comms (IN) Cleared	3
Lost Comms (OUT)	4
Lost Comms (OUT) Cleared	5
Earth Fault (OUT)	6
Earth Fault (OUT) Cleared	7
NW Display Lost	8
NW Display Lost Cleared	9
NW Card Missing	10
NW Card Missing Cleared	11
Panel Connection Comms Error	12
Panel Connection Comms Error Cleared	13
Panel RS232 Connection Error	14
Panel RS232 Connection Error Cleared	15
Door Interlock Fault	16
Door Interlock Fault Cleared	17
Panel Type Fault	18
Panel Type Fault Cleared	19
Panel Country Fault	20
Panel Country Fault Cleared	21
Network Message Checksum Error	22
Network Message Checksum Error Cleared	23
Program/Config CRC Error	24
Program/Config CRC Error Cleared	25
Graphics System Comms Error	26
Graphics System Comms Error Cleared	27

Table 9.2: Network Error Log Codes

Alternatively, a program such as HyperTerminal can be used to access the panel network card Fault log.

Set the parameters to a baud rate of 9600, 8 data bits, No parity, and 1 Stop bit using the on-board RS232 port. No Flow control is necessary.

Commands available with HyperTerminal are:

ver	displays the firmware version of the network card
rst (ENTER)	displays the **Loop Out No Reply** Error Counter
log (ENTER)	displays the first 16 entries of the 128 events in the error log
(ENTER)	while in log access mode, displays the subsequent 16 entries in the error log
(ESC)	exits error log access mode

10. NETWORK SYSTEM TECHNICAL SPECIFICATION

Network Communications Type	RS485
Network Communications Baud Rate	19,200 bps (version 2.00), 38400 bps (version 3.00)
Network Communications Protocol	Pertronic Proprietary
Standard Network Cable Type	Pertronic EVACCBL
Network Cable Specification (EVACCBL)	1.5mm^2 twisted pair (16.6 ohms per km.)
Optional Network Cable Type	Screened twisted pair (for use in potentially electrically noisy environments)
Maximum Network Cable Length	1000m per link (between network cards) using 1.5mm^2 conductors
Network Cable Terminator Resistance	None - included in Network Card hardware
Standard RS485 Cable Type **- to NDU, NCU or Network LED mimic**	Pertronic SGDCI or SGDCE
RS485 Cable Specification (SGDCI internal)	Data: 0.5mm^2 twisted pair. 87.8 ohms per km. Power: 0.8mm^2 pair 34.3 ohms per km.
RS485 Cable Specification (SGDCE external)	Data: 0.5mm^2 twisted pair. 87.8 ohms per km. Power: 1.13mm^2 pair 18.1 ohms per km.
Network Card Addressing	1 to 99 inclusive
Maximum Network Cards	99
Network Card Current Consumption	54mA at 27Vdc (PCB v2.09), 75mA at 27Vdc (PCB v2.20)
Network Control Units per Network Card	1
Network Control Unit Addressing	1
Network Display Units / Mini-Mimics per Network Card	8
Network Display Unit /Mini-Mimic Addressing	0 to 7 inclusive
Network LED Mimics per Network Card	8
Network LED Mimic Addressing	1 to 8 inclusive
RS232 Port Settings	9600:8:N:1 (No Flow control)
Network Display Unit (NDU)	
Dimensions: H x W x D (mm)	200 x 375 x 50
Current Consumption	Quiescent: 6mA @ 27Vdc Alarm: 44mA @ 27Vdc
Network Control Unit (NCU) – stand-alone	
Dimensions: H x W x D (mm)	232 x 375 x 50
Current Consumption	Quiescent: 15mA @ 27Vdc Alarm: 37mA @ 27Vdc

Table 10.1: Technical Data

Document Change History

Issue Number	Reason for Update	Description of Changes	Author
Issue 1: 05/2006	Initial Release		DerekW
Issue 2 08/2006	Network card (1.06) and Utility (2.06) software upgrades	Network time and date synchronisation, read and verify configuration and several other minor changes	DerekW
Issue 3 06/2007	Network Card (2.00) and Utility (2.08) software upgrades. Network Pager bases.	Added extra error messages, re-format of NCU and NDU displays, 32 character descriptors, support for 159 detector addresses, pager and supervisory configuration option, mini-mimic support, NCU simulator added, version compatibility checking in Utility Section 6: Pager protocol details added.	DerekW GeoffT
Issue 4 09/2008	Network Card (3.00) and Utility (2.08) software upgrades.	New hardware version v2.20 for use with version 3.0 firmware. Added history log menu to NCU / NDU, and utilities download function for history log events. Printer interface added. Mini Mimics added. Updated for new utility features and changes including network mapping by card and network diagnostic sections. Changed maximum cards limit to 99.	Daniel JohnF
Issue 5 04/09	Network Card (2.04 and 3.04) and Utility (2.10) software upgrades	Added Printer/Pager daily test messages function	Daniel DerekW
Issue 6 05/11	Update for 2.08 software, including new functionality.	Updated text, added key section, and replaced technical diagrams.	ChrisB
Issue 6.1 Nov 2012	Cross references	Re-established cross references section 8	RDB
Issue 6.2 Oct 2014	CN 1863	Changes to Kew pager format section 6.4	RDB